

Data visualization and modelling

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4-Non-parametric Bootstrap

Overview

- 1- The method
- 2- Some methods to calculate CI
- 3- Examples of application

The bootstrap method (parametric or non-parametric) has three stages:

- 1- Estimate the population (or probability) distribution from the data set.
- 2- Simulate the sampling from the population distribution obtaining the new bootstrap samples.
- 3- For each sampling, calculate the sample statistic of interest.

For the non-parametric Bootstrap, we simply use the frequency distribution of the n data values as our best guess of the population or probability distribution. In other words the list of original values is assumed to be our population distribution. Clearly this will be an increasingly poor estimation the fewer the observations.

THE NON-PARAMETRIC BOOTSTRAP APPROACH

Resampling methods: a practical guide to data Analysis. Phillip I. Good (2006). Chapter 2, p.8.

2.2.1 The Bootstrap

One practical alternative, known as the *bootstrap*, is to treat the original sample of values as a stand-in for the population and to resample from it repeatedly, *with replacement*, computing the desired estimate each time.

Consider the following set of 22 observations on the heights of sixth-grade students, measured in centimeters and ordered from shortest to tallest. Note that the median of the sample, a plug-in estimate of the median of the population from which it is drawn, is 153.5 cm.

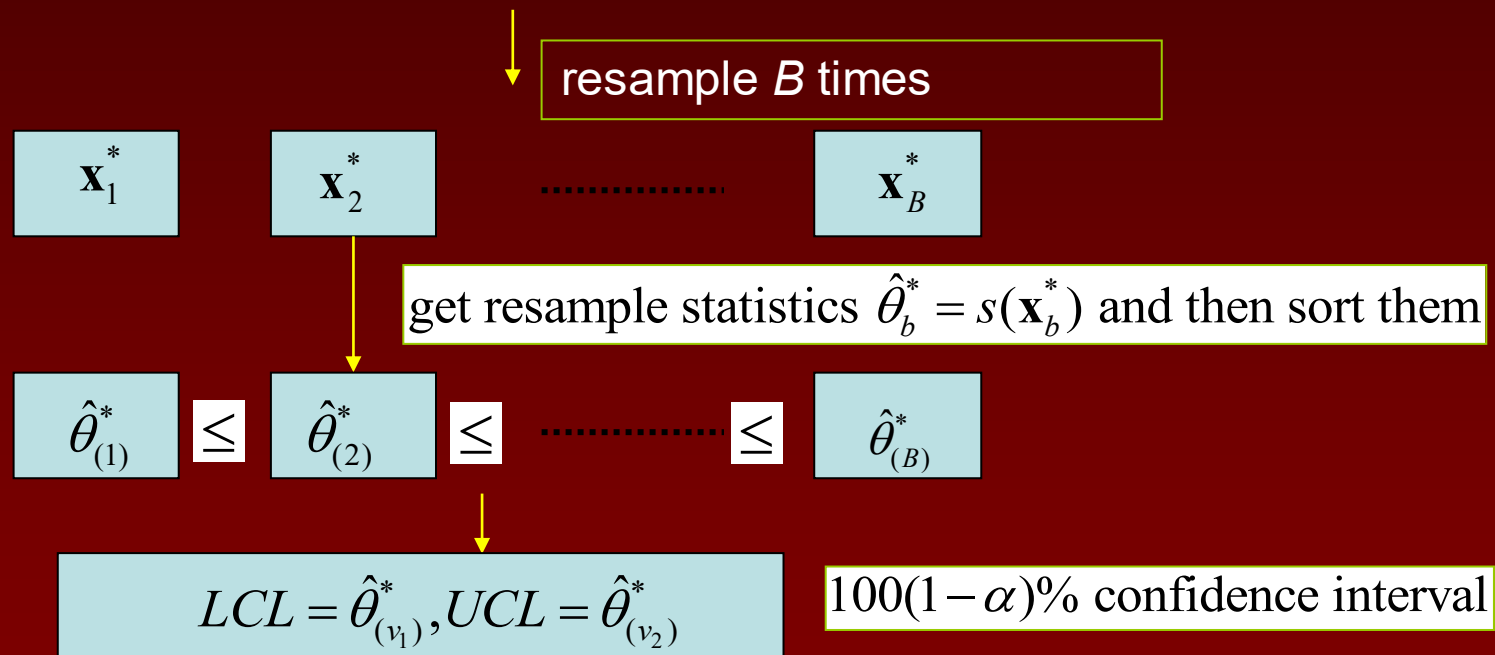
137.0	138.5	140.0	141.0	142.0	143.5	145.0	147.0	148.5	150.0	153.0
154.0	155.0	156.5	157.0	158.0	158.5	159.0	160.5	161.0	162.0	167.5

A simple example: estimation and CI for the median.

```
class=c(141,156.5,162,159,157,143.5,154,158,140,142,150,148.5,  
138.5,161,153,145,147,158.5,160.5,167.5,155,137)  
n = length(class)  
#set number of bootstrap samples  
nsim=1000  
stat = numeric(nsim) #create a vector in which to store the results  
#Set up a loop to generate a series of bootstrap samples  
for (i in 1:nsim){  
  classB= sample (class, n, replace=T)  
  stat[i] = median(classB)}  
boxplot (stat)  
hist(stat)
```


Flowchart of the Percentile or Quantile method to calculate confidence intervals

$$\text{data } \mathbf{x} = (x_1, x_2, \dots, x_n) \Rightarrow \hat{\theta} = s(\mathbf{x})$$



Here v_1 and v_2 are the orders of the $\alpha/2$ and $1 - \alpha/2$ quantiles

The percentile method to calculate confidence intervals

Resampling methods: a practical guide to data Analysis. Phillip I. Good (2006). Chapter 2, p.18.

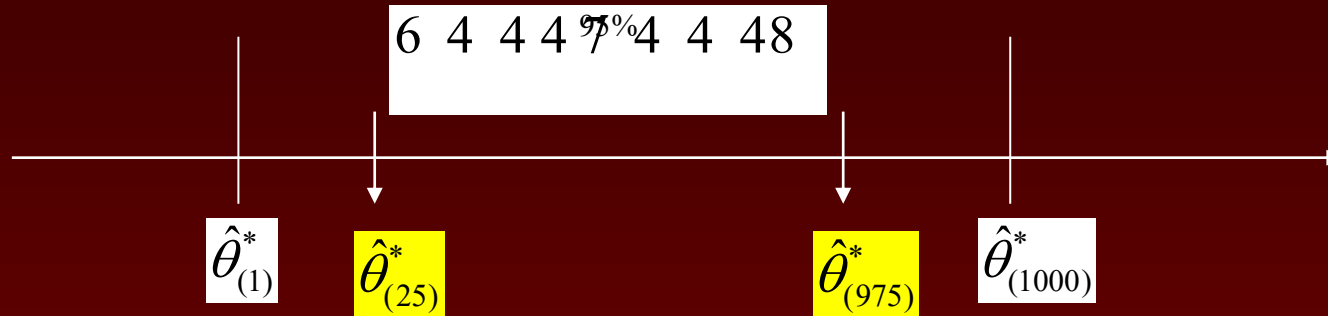
1. Is it accurate? That is, is it correct to say that 95% of the time we expect this interval to cover the true value of the population parameter?

2. Is it efficient? That is, is the interval no wider than it should be, so that the probability it includes false values of the population parameter is as small as possible?

Unfortunately, the answer for the percentile bootstrap confidence interval in both cases is “No.”

Percentile bootstrap intervals are good when the estimate is symmetrically distributed about the true value of the parameter and the tails of the estimate's distribution drop off rapidly to zero.

The basic or simple method



$$\therefore P_* \left\{ \hat{\theta}_{(25)}^* \leq \hat{\theta}^* \leq \hat{\theta}_{(975)}^* \right\} \approx 1 - \alpha = 95\%$$

$$= P_* \left\{ \hat{\theta}_{(25)}^* - \hat{\theta} \leq \hat{\theta}^* - \hat{\theta} \leq \hat{\theta}_{(975)}^* - \hat{\theta} \right\}$$

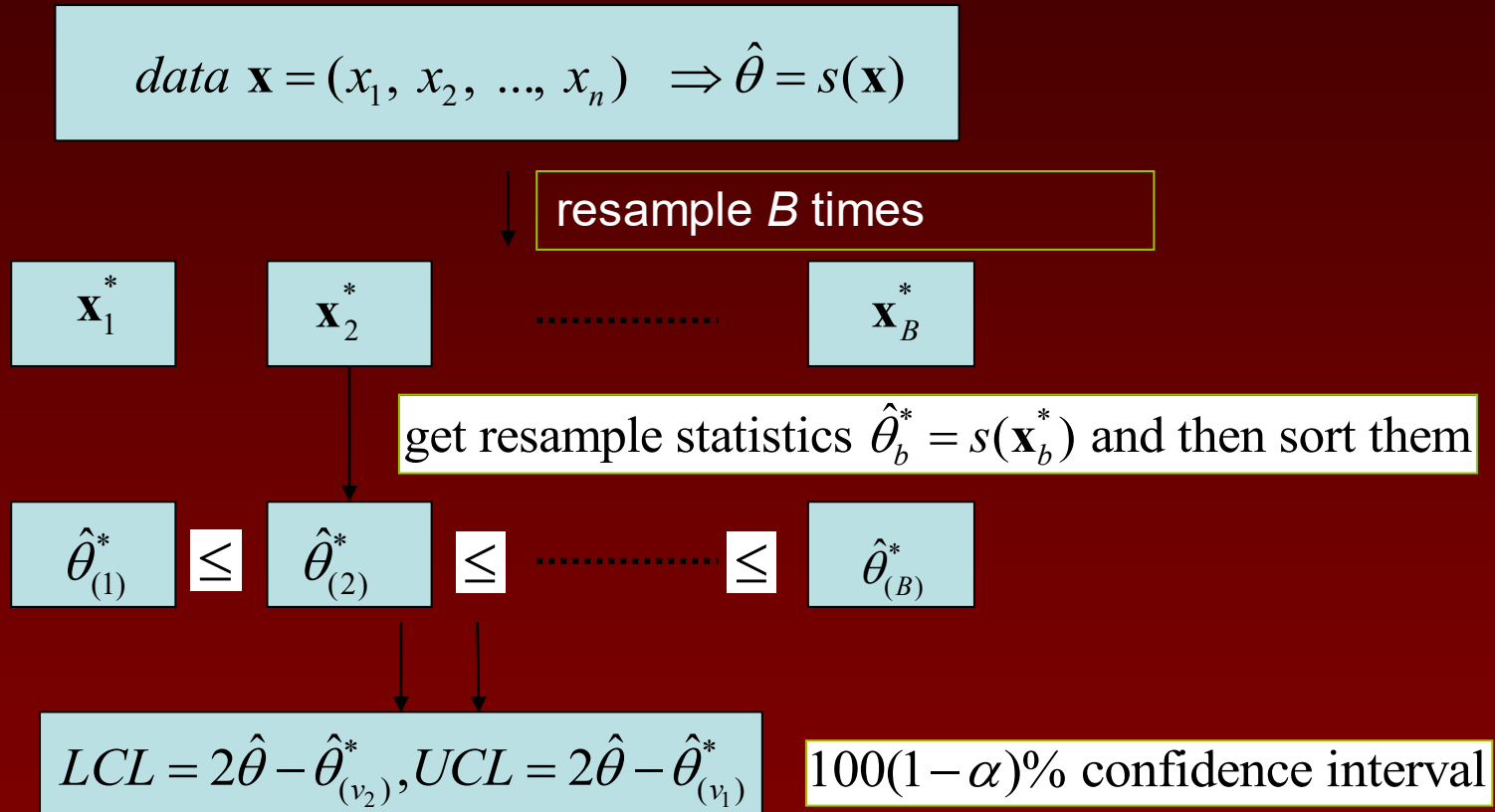
$$\approx P \left\{ \hat{\theta}_{(25)}^* - \hat{\theta} \leq \hat{\theta} - \theta \leq \hat{\theta}_{(975)}^* - \hat{\theta} \right\}$$

$$= P \left\{ 2\hat{\theta} - \hat{\theta}_{(975)}^* \leq \theta \leq 2\hat{\theta} - \hat{\theta}_{(25)}^* \right\}.$$

$$\therefore [LCL = 2\hat{\theta} - \hat{\theta}_{(975)}^*, UCL = 2\hat{\theta} - \hat{\theta}_{(25)}^*]$$

is an approximate $(1 - \alpha)$ confidence interval for θ .

Flowchart of the Simple Method



Here v_1 and v_2 are the orders of the $\alpha/2$ and $1 - \alpha/2$ quantiles

The example for the median.

95% Simple Confidence Interval:

$2 * \text{median}(\text{class}) - \text{quantile}(\text{stat}, 0.975)$ 149.5

$2 * \text{median}(\text{class}) - \text{quantile}(\text{stat}, 0.025)$ 162.0 155.75 ± 6.25

Note that the length of this interval is just the same than the percentile confidence interval.

The bootstrap-t method to calculate confidence intervals

Assume that we have consistent estimates of θ_0 and its standard error at hand and that the asymptotic distribution of the t-statistic is the standard normal.

$$t = \frac{\hat{\theta} - \theta_0}{\widehat{se}(\hat{\theta})} \xrightarrow{d} N(0, 1)$$

Draw B independent bootstrap samples of size n from the original sample. Estimate the t-value for each sample:

$$t_b^* = \frac{\hat{\theta}_b^* - \hat{\theta}}{\widehat{se}_b^*(\hat{\theta})} \quad \text{for } b = 1, \dots, B$$

where $\hat{\theta}_b^*$ and $\widehat{se}_b^*(\hat{\theta})$ are estimates of the parameter θ and its standard error using the bootstrap sample.

Calculate $t_{\alpha/2}$ and $t_{1-\alpha/2}$ by means of the quantiles of t_b^* . Finally, the CI is constructed as follows,

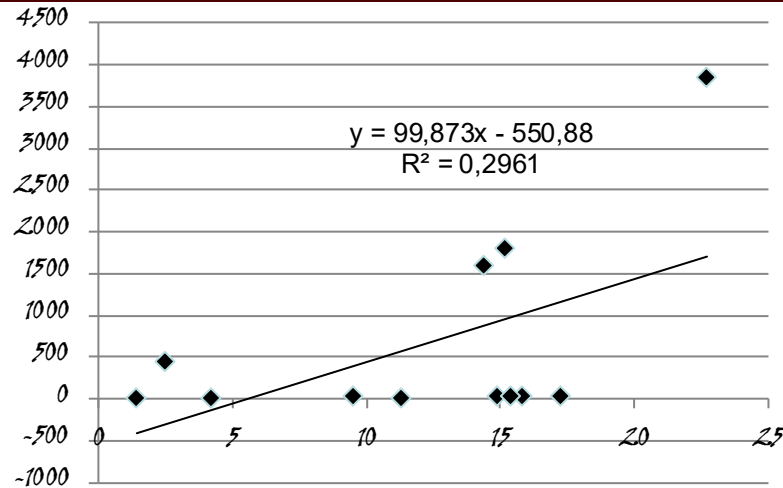
$$[\hat{\theta} + t_{\alpha/2} \cdot \widehat{se}(\hat{\theta}), \hat{\theta} + t_{1-\alpha/2} \cdot \widehat{se}(\hat{\theta})]$$

Again the example of correlation between gill weight and body weight in 12 crabs

Gill (mg) Body (g)

1590	14,4
1790	15,2
10	11,3
450	2,5
3840	22,7
23	14,9
10	1,41
32	15,81
8	4,19
22	15,39
32	17,25
21	9,52

$r = 0,54417901$



We want to calculate a 95% confidence interval for the slope.

```
gill<-c(1590,1790,10,450,3840,23,10,32,8,22,32,21)  
body<-c(14.40,15.20,11.30,2.50,22.70,14.90,1.41,15.81,  
4.19,15.39,17.25,9.52)
```

```
n<-length(gill); data<-cbind(body,gill)
```

```
fit<-lm(data[,2]~data[,1])
```

These are the parameter estimate and the standard error

```
theta<-summary(fit)$coefficients[2]
```

```
sdtheta<-summary(fit)$coefficients[4]
```

```
> summary(fit)$coefficients
```

	Estimate	S td. Error	t value	Pr(> t)
(Intercept)	-550.88200	659.90900	-0.8347848	0.42333335
data[, 1]	99.87262	48.69122	2.0511424	0.06737933

```
nb<-1000; y<-seq(1,n);tb<-numeric(nb)
```

```
for(i in 1:nb){
```

```
  yb<-sample(y,n,replace=T)
```

```
  fitb<-lm(data[yb,2]~data[yb,1])
```

```
  thetab<-summary(fitb)$coefficients[2]
```

```
  sdthetab<-summary(fitb)$coefficients[4]
```

```
  tb[i]<-(thetab-theta)/sdthetab}
```

And this is the bootstrap-t 95% CI:

```
theta+sdtheta*quantile(tb,0.025)
```

-376.3289

```
theta+sdtheta*quantile(tb,0.975)
```

215.9330

We can compare it with the quantile method CI,

```
nb<-1000; y<-seq(1,n);thetab<-numeric(nb)
```

```
for(i in 1:nb){
```

```
yb<-sample(y,n,replace=T)
```

```
fitb<-lm(data[yb,2]~data[yb,1])
```

```
thetab[i]<-summary(fitb)$coefficients[2] }
```

```
quantile(thetab,c(0.025,0.975))
```

2.5% 97.5%

-7.001839 228.493311

On the statistical analysis of ratios. An example of bootstrap application

The 2D:4D index.



Hands are present in many cave paintings.



Cave “las Manos”, Argentina



Cave of Gargas, french Pyrenees

Some images are positive, others are negative

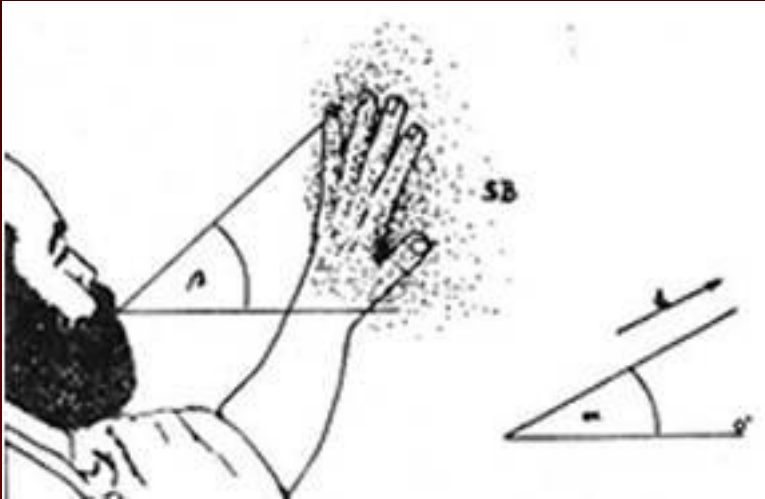


Cave of Chauvet, France



Cave of “Fuente del Salín”, Cantabria

The negative images have a higher anthropometric quality than the positives.



Several experiments have been performed in order to understand how the hand prints were made.



Is it possible to know the sex of a person by means of the hand prints ?

This new appeared in the newspapers (2006):

SCIENCE ACTUALITÉS.fr

LE MAGAZINE QUI SE VISITE AUSSI À LA CITÉ DES SCIENCES

ENQUÊTES

ARCHÉOLOGIE & PALÉONTOLOGIE

Empreintes préhistoriques : histoire de sexes à Bornéo

Grâce à un logiciel inédit, une équipe du CNRS a pu déterminer l'appartenance sexuelle des empreintes de mains retrouvées dans une grotte préhistorique de Bornéo. Ces résultats ouvrent de nouvelles perspectives dans l'interprétation des peintures rupestres. À Bornéo, mais aussi dans le reste du monde.

Olivier Boulanger, le 23/01/2006

Découverte en 1998 par l'ethno-archéologue Jean-Michel Chazine et son équipe*, la grotte préhistorique [Gua Masri II](#) sur l'île de Bornéo est un lieu énigmatique. Il y a quelque 10 000 ans, peut-être plus, nos ancêtres sont venus laisser ici leurs empreintes sur les parois. Mais à la différence de nombreuses autres grottes, celle-ci ne laisse apparaître aucune représentation figurative, pas d'animaux, ni de scènes de chasse : seulement des mains dites « négatives », des dizaines d'empreintes dont l'organisation suggère qu'elles n'ont pas été placées au hasard. Un cas unique dans le monde, selon son découvreur. Depuis quelques semaines, Jean-Michel Chazine estime être en mesure d'apporter de nouveaux éléments sur ces étranges témoignages : ils ne sont pas seulement l'œuvre d'hommes préhistoriques, mais aussi de femmes !

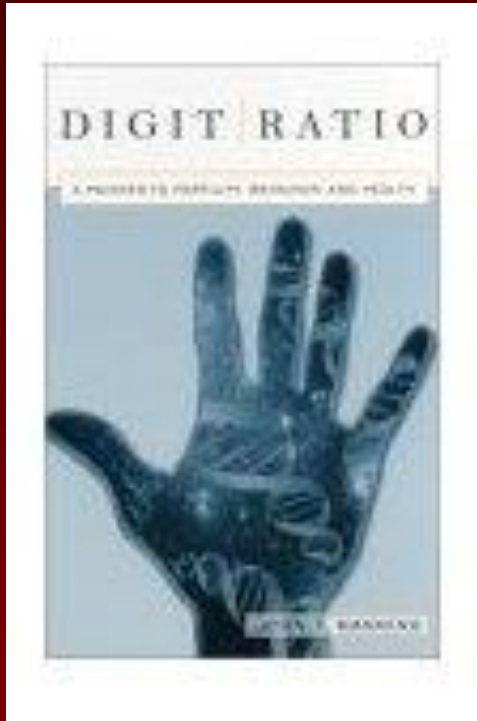
Empreintes négatives, empreintes positives...

Les hommes préhistoriques ont utilisés principalement deux méthodes pour laisser leurs empreintes de main.

La première consiste à appliquer la main sur la paroi et à projeter un pigment en soufflant avec la bouche. La main apparaît alors en négatif.

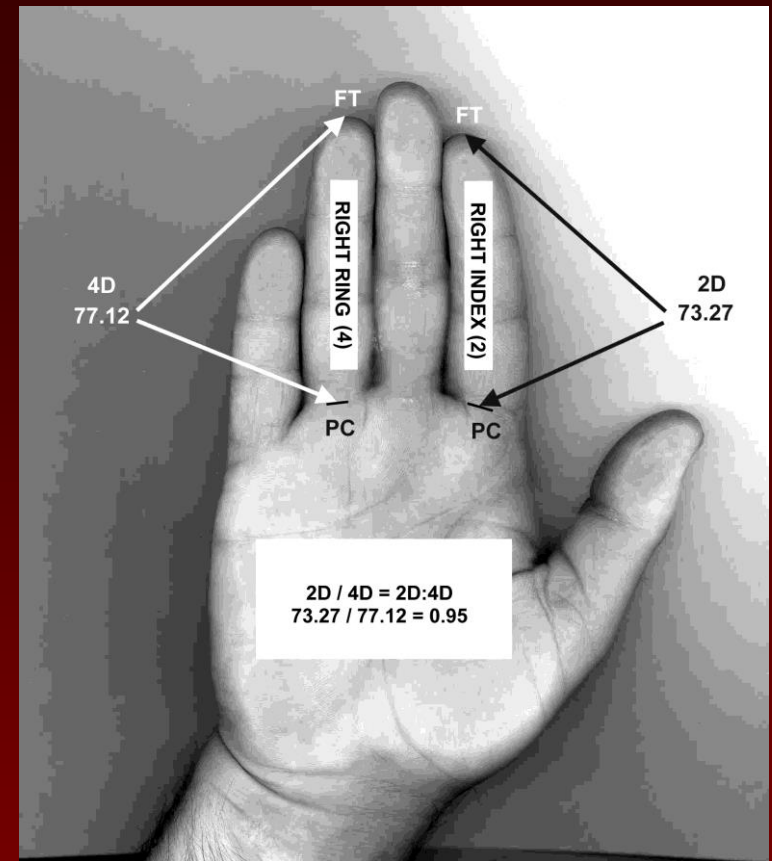
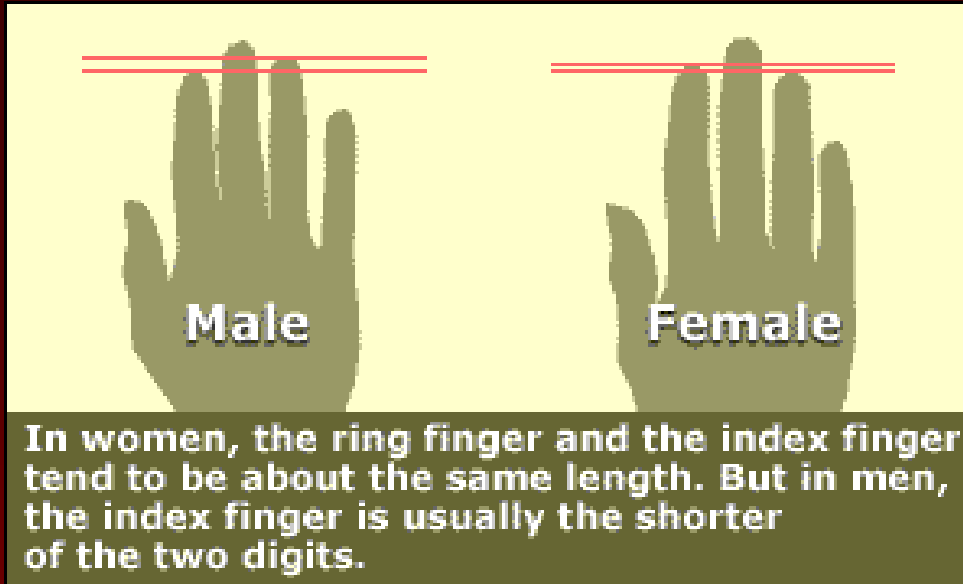
La seconde méthode consiste à tremper la main directement dans le pigment et à l'apposer sur la paroi. L'empreinte est alors positive.

About Manning Index



**John T. Manning. Department of Psychology
University of Central Lancashire**

Digit Ratio: A Pointer to Fertility, Behavior and Health
by John T. Manning
NJ: Rutgers University Press. 2002



Manning index is the ratio between the lengths of the second and the fourth fingers. It is also called 2D:4D ratio.



Does this hand belong to a man or a woman?

Project work in Statistics. Academic year 2005-2006.

Measurement protocol:

The students measured 2D and 4D of both hands using a caliper.
The lengths were taken from the major crease at the base of the digit to the tip.

It is interesting to repeat the measures several times.

Measures are different if they are taken from photocopies.
Don't mix measures coming from different protocols!



Project work in Statistics. Academic year 2005-2006.

The measures were collected and put in an Excel worksheet.

	A	B	C	D	E
1	Individu	Sexe	IM Dreta	IM Esquerra	
2	1	h	0,96	0,97	
3	2	d	0,99	0,98	
4	3	h	0,97	0,96	
5					

- Graphical representation
- Paired T-test
- Two sample T-test
- Lineal regression
- etc...

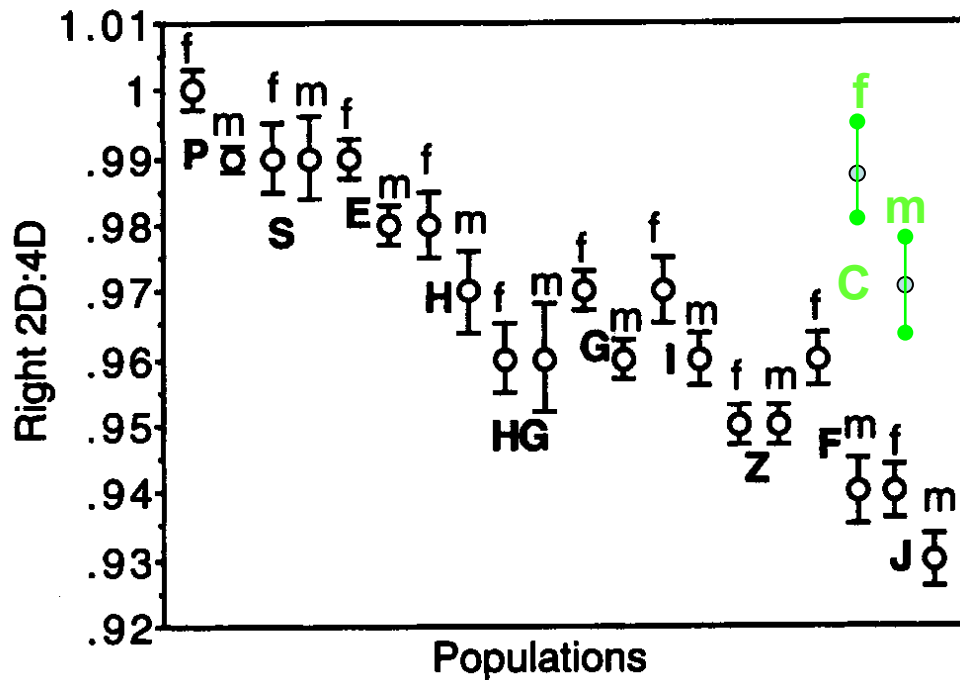


Figure 1.7. Differences in 2D:4D ratio of the right hand between populations and sexes in 10 cultures. P = Poland, S = Spain, E = England, H = Hungary, HG = Hungary Gypsy, G = Germany, I = India, Z = Zulu (South Africa), F = Finland, J = Jamaica. Sex is indicated by f = female and m = male. Substantial between-population differences are revealed. The Polish, Spanish, and English samples show high 2D:4D ratios; the German, ethnic Hungarian, Hungarian Gypsies, and Indians intermediate values of 2D:4D; the Zulu, Finnish, and Jamaican samples show low 2D:4D ratios. The mean ratios are rounded to the nearest 0.01. **C = Catalonia**

Why this sexual dimorphism?

“The relative lengths of the 2nd and 4th digits are influenced prenatally by testosterone and estrogen concentrations...” (Manning, 2002)

More than 700 papers are recorded by the ISI Web of Knowledge containing 2D:4D in their titles.

1. Title: **Your fate is in your hands? Handedness, digit ratio (2D:4D), and selection to a national talent development system**
Author(s): Baker, Joseph; Kungl, Ann-Marie; Pabst, Jan; et al.
Source: LATERALITY Volume: 18 Issue: 6 Pages: 710-718 DOI: 10.1080/1357650X.2012.755992 Published: DEC 1 2013
Times Cited: 0 (from All Databases)
 [[View abstract](#)]
2. Title: **A Critical Review of the Research on the Extreme Male Brain Theory and Digit Ratio (2D:4D)**
Author(s): Teatero, Missy L.; Netley, Charles
Source: JOURNAL OF AUTISM AND DEVELOPMENTAL DISORDERS Volume: 43 Issue: 11 Pages: 2664-2676 DOI: 10.1007/s10803-013-1819-6
Published: NOV 2013
Times Cited: 0 (from All Databases)
 [[View abstract](#)]
3. Title: **2D:4D finger length ratio and reproductive indices in a Chuvashian population**
Author(s): Kalichman, Leonid; Batsevich, Valery; Kobylansky, Eugene
Source: AMERICAN JOURNAL OF HUMAN BIOLOGY Volume: 25 Issue: 5 Pages: 617-621 DOI: 10.1002/ajhb.22420 Published: SEP 2013
Times Cited: 0 (from All Databases)
 [[View abstract](#)]

Why the 2D:4D ratio?

- The empirical fact that the hands in women are more symmetric than in men is not a new discovery.

Ecker, A. (1875). Einige Bemerkungen tiber ein schwankenden Charakter in der Hand des Menschen. *Arch. Anthrop. Brnschw*, 8, p. 67-74.

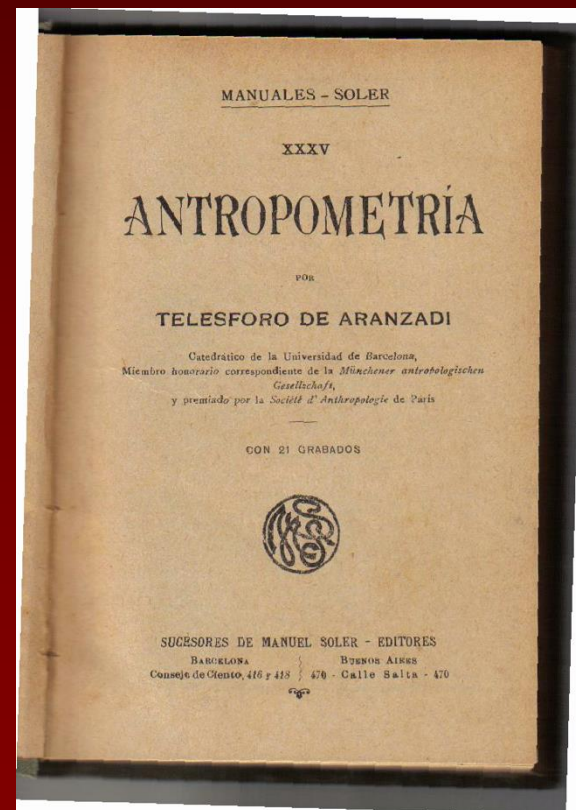
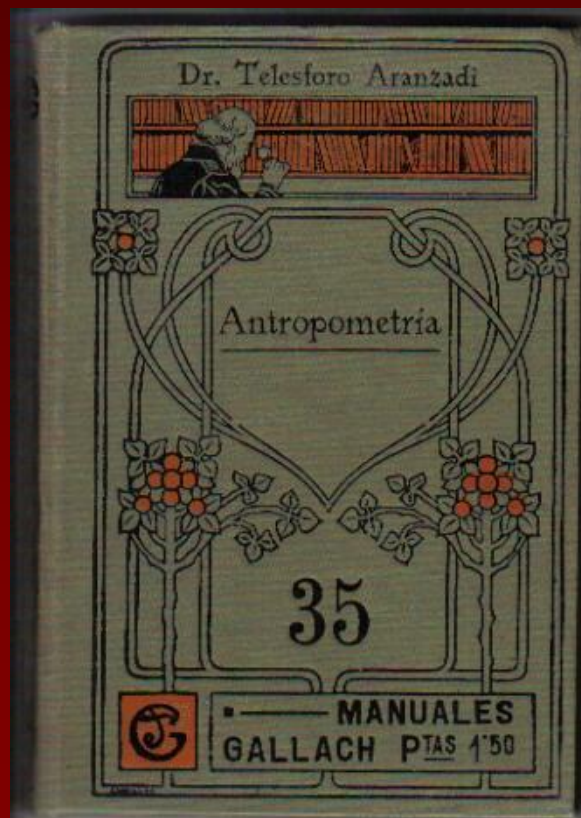
Baker, F. (1888). Anthropological notes on the human hand. *American Anthropologist*, 1, p. 51-76.

Phelps, V.R. (1952). Relative index finger length as a sex-influenced trait in man, *Am. J. Hum. Genet.*, 4, p. 72–89.

infantiles ó de menor desarrollo. Las labradoras letas ó lituanias que trabajan en la heredad tienen mayor desarrollo mecánico y se diferencian menos del hombre (1).

Es más frecuente en la mujer el tener el dedo índice más largo que el anular (35 por 100, en vez de 4 por 100), mientras que lo inverso ocurre en el hombre: el pulgar de la mano y el dedo gordo del pie son más cortos en la mujer que en el hombre, y son también más cortas que en éste las segundas falanges de los dedos de los pies.

El peso de la recién nacida es, por término me-



Coming back to the caves



Cave Gua Masri II, Borneo



Détermination de l'identité sexuelle des auteurs des mains de Masri II grâce au logiciel *Kalimain* 1.0 (Newsletter: Chacine, JC and Noury A., 2005).

Is this classification reliable ?

- J. Manning has questioned these results.

Nelson, EC, Manning JT and Sinclair A. (2006).

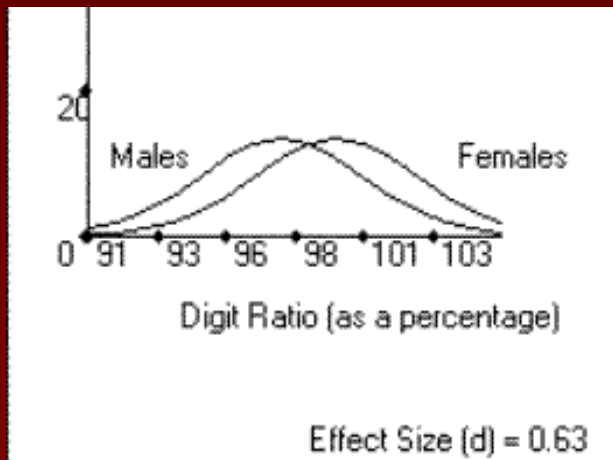
News using the length of the 2nd and 4th digit ratio (2D:4D) to sex cave art hand stencils: factors to consider. Before Farming, 1, A6.

- The values of the 2D:4D ratio change according to the ethnic group of the individuals.
- Chazin and Nouri use the average of the European populations (0.96 for men and 1.0 for women).
- Left or right hands produce different values.
- The transition between real hand and hand print has many factors of error that have not been studied.

The distribution of 2D:4D ratio

Most researchers agree to assume normality for these kind of data sets. Moreover, goodness of fit tests also agree with this assumption.

A measure of the distance between populations according to the 2D:4D values is the “effect size statistic”:



$$\hat{d} = \frac{\bar{X}_d - \bar{X}_h}{S_p}$$

However the CLESS is more intuitive

CLESS: Common language effect size statistic

McGraw and Wong (1992), *Psychological Bulletin*, 111, p. 361-365

CLESS is an estimation of the probability that a randomly selected value from the population with the greatest expectation will be greater than a randomly sampled value from the other population.

For our situation, $CLESS \approx P(X_d - X_h > 0) = P(Z < \frac{\mu_d - \mu_h}{\sigma\sqrt{2}}) = \Phi(d / \sqrt{2})$

For instance, using the data from Liverpool (Manning),

$d = 0.63$

$CLESS = 0.67$

and for the data collected by our students,

$d = 0.62$

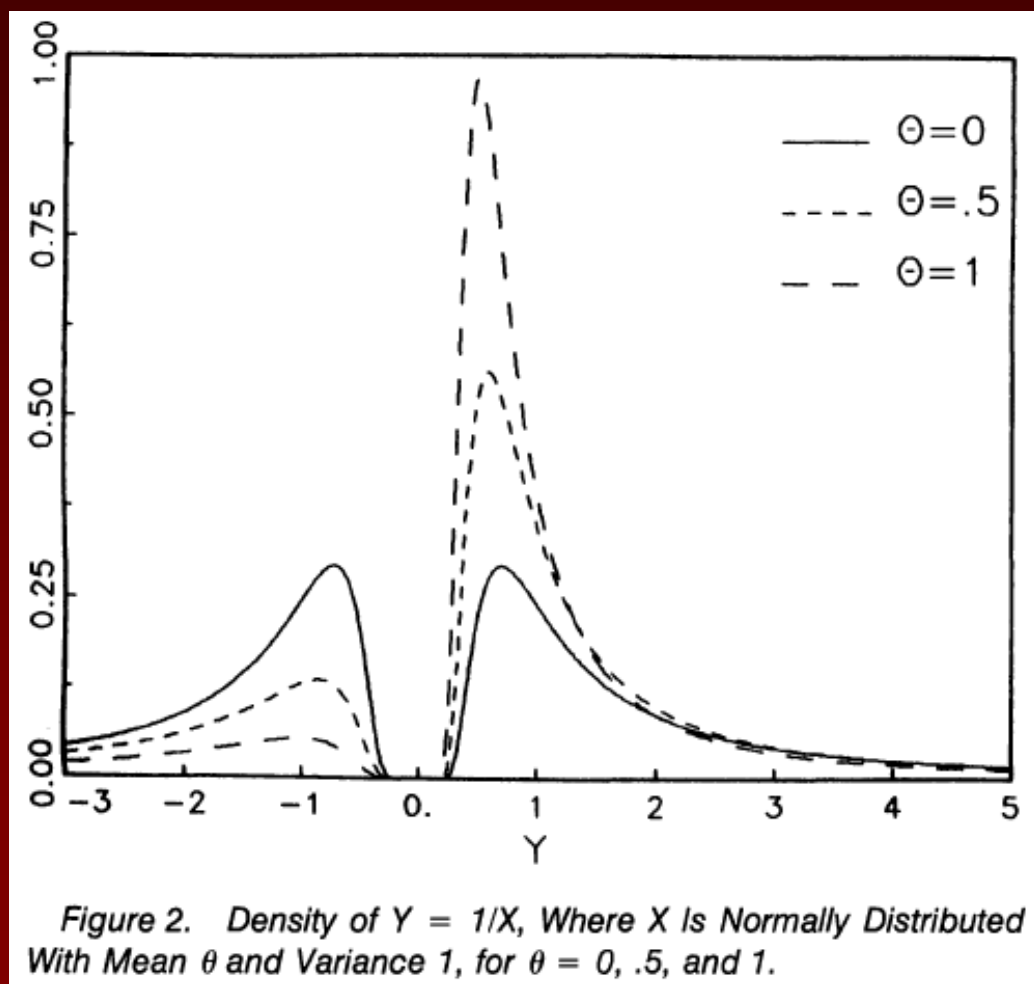
$CLESS = 0.67$

Can normality be assumed?

The quotient 2D:4D is in some sense arbitrary.
Why not consider 4D:2D instead?

It is natural to expect that the same statistical model used for 2D:4D could also describe the 4D:2D ratios, that is, the data set of the reciprocals.

However, if the r.v. X is normally distributed....
... $1/X$ is not normally distributed !!



Lehmann and Popper (1988)
Inverted Distributions.
Am. Stat., 42, p. 191-194.

We are going to analyze our data set using several methods:

```
iw<-c(0.998,0.950,1.023,0.980,0.907,0.958,1.016,0.957,0.998,  
0.979,0.982,1.000,0.998,1.017,0.976,0.967,1.023,1.033,  
0.979,0.983,0.990,0.998)
```

```
im<-c(0.931,1.019,0.936,0.974,0.949,0.939,0.984,0.956,0.947,  
0.969,0.979,0.922,0.973,1.014,0.936,0.966,1.020,0.975,1.015)
```

```
t.test(iw,im)
```

data: iw and im

t = 1.9423, df = 37.058, p-value = 0.05973

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-0.00078851 0.03734353

sample estimates:

mean of x mean of y

0.9869091 0.9686316

Three kind of bootstrap CI for the difference of means:

```
nw<-length(iw); nm<-length(im);nb<-1000
```

```
difb<-numeric(nb)
```

```
for(i in 1:nb){
```

```
  iwb<-sample(iw,nw,replace=T)
```

```
  imb<-sample(im,nm,replace=T)
```

```
  difb[i]<-mean(iwb)-mean(imb)}
```

#Quantile method:

```
quantile(difb,c(0.025,0.975))
```

2.5% 97.5%

0.001472727 0.035580323

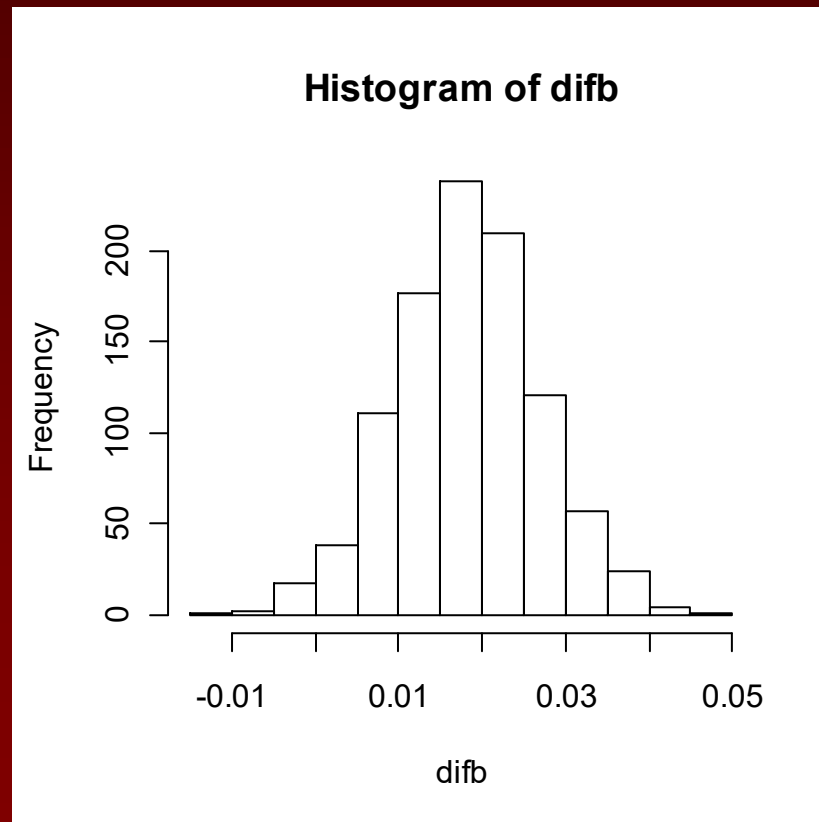
#Simple method:

$2 * (\text{mean}(\text{iw}) - \text{mean}(\text{im})) - \text{quantile}(\text{difb}, 0.975)$

0.000974701

$2 * (\text{mean}(\text{iw}) - \text{mean}(\text{im})) - \text{quantile}(\text{difb}, 0.025)$

0.0350823



Bootstrap-t CI method:

```
tb<-numeric(nb)
theta<-mean(iw)-mean(im)
sdtheta<-sqrt(var(iw)/nw+var(im)/nm)
for(i in 1:nb){
  iwb<-sample(iw,nw,replace=T)
  imb<-sample(im,nm,replace=T)
  thetab<-mean(iwb)-mean(imb)
  sdthetab<- sqrt(var(iwb)/nw+var(imb)/nm)
  tb[i]<-(thetab-theta)/sdthetab}
```

And this is the bootstrap-t 95% CI:

```
theta+sdtheta*quantile(tb,0.025)
```

0.0006838654

```
theta+sdtheta*quantile(tb,0.975)
```

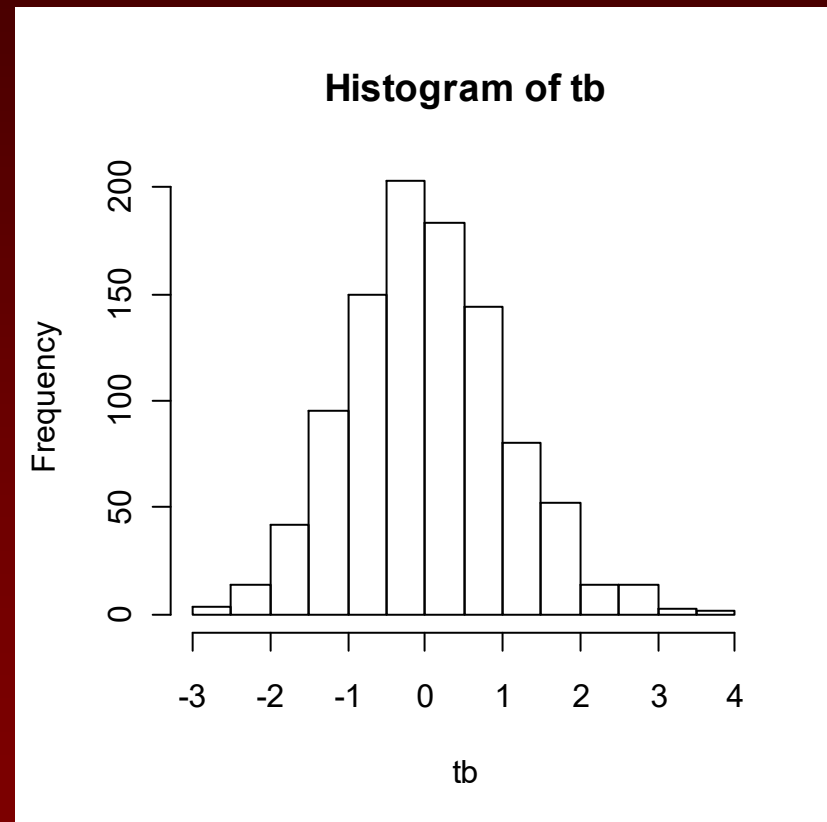
0.03888483

Note the values of the quantiles
of tb:

```
quantile(tb,c(0.025,0.975))
```

2.5% 97.5%

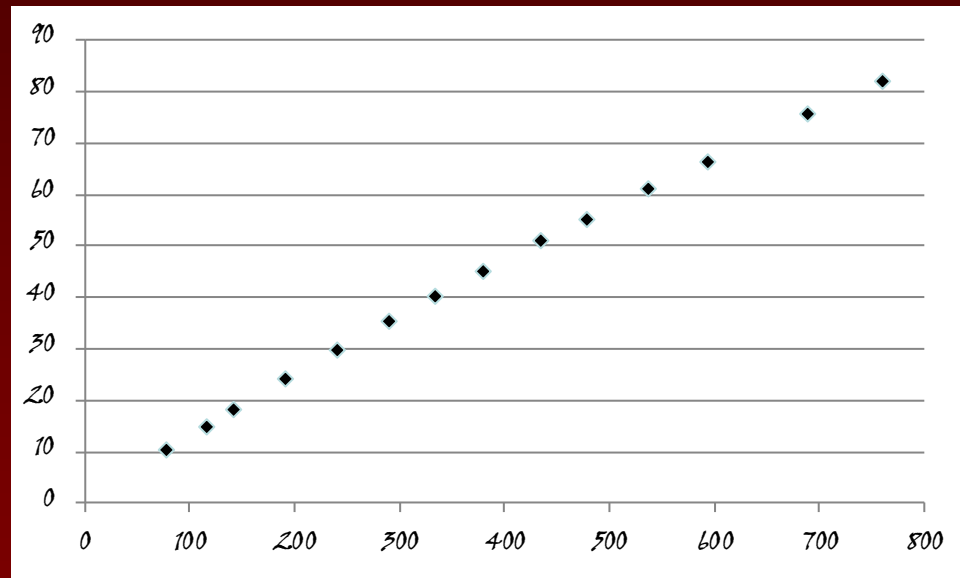
-1.869622 2.189876



The bootstrap method in non-linear regression

These data are the result of a NIST study involving dental research in monomolecular adsorption. The response variable is volume, and the predictor variable is pressure.

y	x
10.07	77.6
14.73	114.9
17.94	141.1
23.93	190.8
29.61	239.9
35.18	289.0
40.02	332.8
44.82	378.4
50.76	434.8
55.05	477.3
61.01	536.8
66.40	593.1
75.47	689.1
81.78	760.0



We can fit this data set by means of the non-linear regression model, $y_i = b_1(1-\exp[-b_2x_i]) + e_i$

```
x<-c(77.6,114.9,141.1,190.8,239.9,289,332.8,378.4,434.8,477.3,536.8,593.1,689.1,760)
y<-c(10.07,14.73,17.94,23.93,29.61,35.18,40.02,44.82,50.76,55.05,61.01,66.4,75.47,81.78)
```

```
ajust<- nls(y ~ b1* (1-exp(-b2*x)), start = list(b1= 2, b2=0.001))
```

```
summary(ajust)
```

```
Formula: y ~ b1 * (1 - exp(-b2 * x))
```

```
Parameters:
```

	Estimate	Std. Error	t value	Pr(> t)
b1	2.389e+02	2.707e+00	88.27	<2e-16 ***
b2	5.502e-04	7.267e-06	75.71	<2e-16 ***

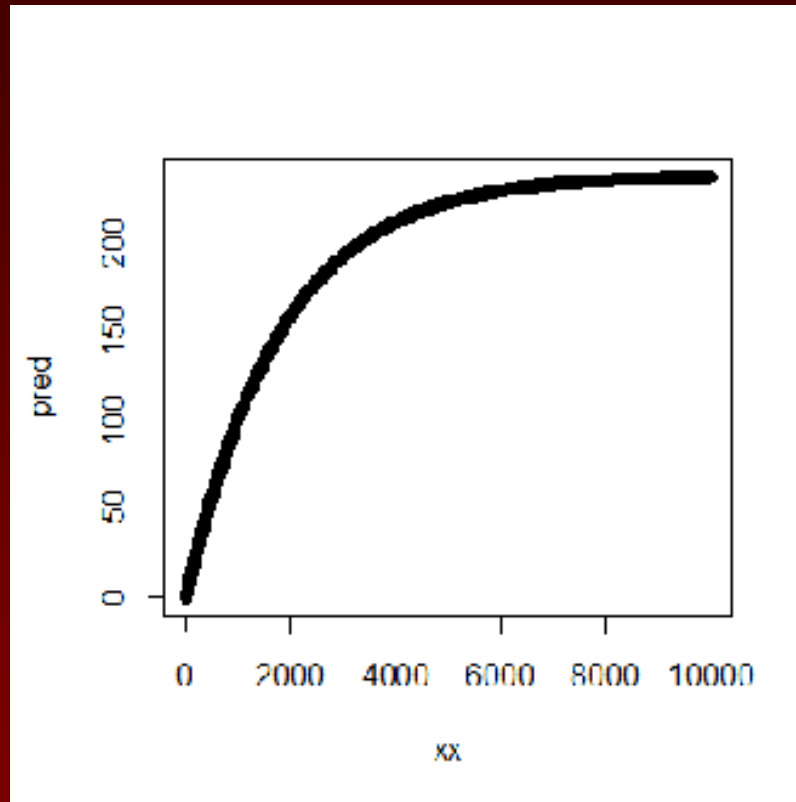
```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1019 on 12 degrees of freedom
```

```
Number of iterations to convergence: 8
```

```
Achieved convergence tolerance: 1.029e-06
```

```
xx<-seq(1,10000)
pred<-summary(ajust)$coefficients[1] * (1-exp(summary(ajust)$coefficients[2] *xx))
plot(xx,pred)
```



We want to calculate a 95% confidence interval for the prediction for $x=900$ and for the maximum value of y (b_1).

We shall consider two methods of bootstrap.

Method I: bootstrapping the pairs (x,y)

For b_1 we shall use the method of the quantiles and the bootstrap-t. For the prediction at $x=900$, only the method of the quantiles will be used.


```
n<-length(x); data<-cbind(x,y)
theta<-summary(ajust)$coefficients[1]
sdtheta<-summary(ajust)$coefficients[3]
nb<-1000; z<-seq(1,n);tb<-numeric(nb);predb<-numeric(nb)
thetab<-numeric(nb)
for(i in 1:nb){
  zb<-sample(z,n,replace=T)
  ajustb<- nls(data[zb,2]~ b1* (1-exp(-b2*data[zb,1])), start =
    list(b1= 2, b2=0.001))
  thetab[i]<-summary(ajustb)$coefficients[1]
  sdthetab<-summary(ajustb)$coefficients[3]
  predb[i]<- summary(ajustb)$coefficients[1] * (1-exp(-
    summary(ajustb)$coefficients[2] *900))
  tb[i]<-(thetab[i]-theta)/sdthetab}
```

And this is the bootstrap-t 95% CI of b_1 :

`theta+sdtheta*quantile(tb,0.025)`

217.1870

`theta+sdtheta*quantile(tb,0.975)`

243.9387

This is the CI based in the quantile method,

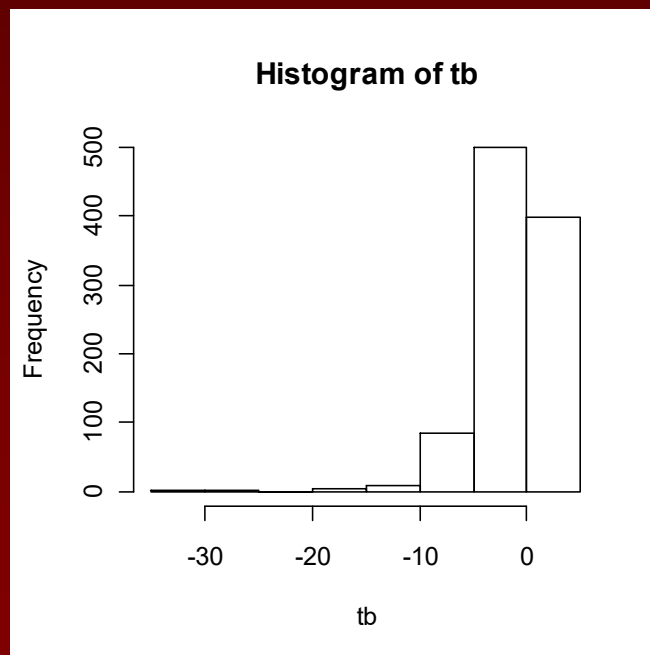
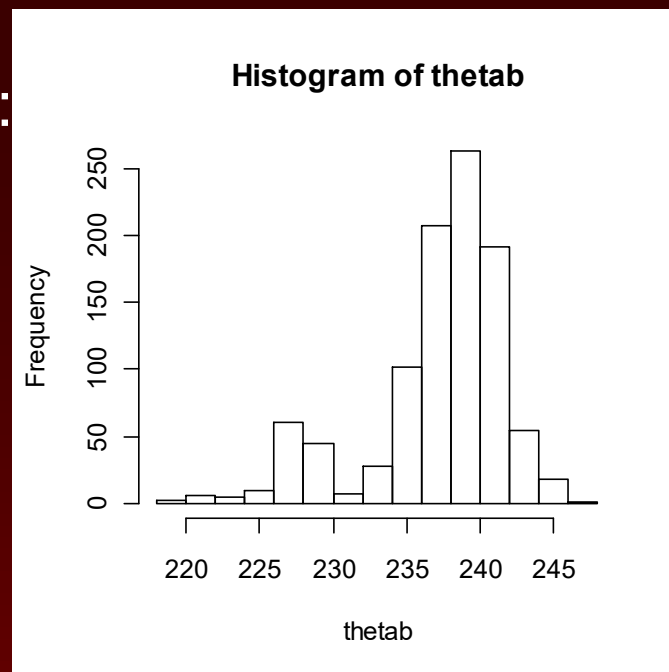
`quantile(thetab,c(0.025,0.975))`

2.5% 97.5%

226.1510 243.4359

The classical CI is,

238.9 ± 5.3

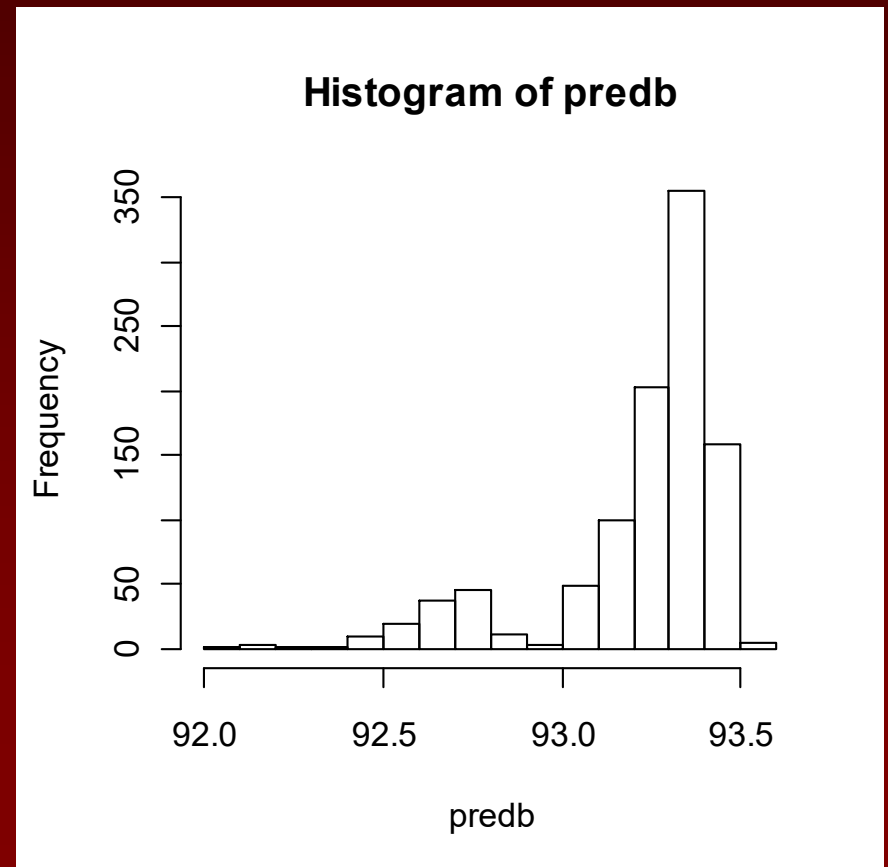


This is the quantile 95% CI of the prediction at x=900:

```
quantile(predb,c(0.025,0.975))
```

2.5% 97.5%

92.56370 93.45463



Method II: bootstrapping the residuals

- 1- Fit the original sample and keep the residuals and the estimates of the parameters.

```
x<-c(77.6,114.9,141.1,190.8,239.9,289,332.8,378.4,434.8,477.3,536.8,593.1,689.1,760)
y<-c(10.07,14.73,17.94,23.93,29.61,35.18,40.02,44.82,50.76,55.05,61.01,66.4,75.47,81.78)
ajust<- nls(y ~ b1* (1-exp(-b2*x)), start = list(b1= 2, b2=0.001))
res<-residuals(ajust)
```

- 2- Construct the bootstrap samples using the original x and calculating y_b as follows,

$$y_{bi} = b^*_1(1-\exp[-b^*_2x_i]) + r_{bi}$$

where b^*_1 and b^*_2 are the estimates of the parameters and r_b is a bootstrap sample of the residuals.

```
n<-length(x); c1<-summary(ajust)$coefficients[1]
c2<- summary(ajust)$coefficients[2]
sdtheta<-summary(ajust)$coefficients[3]
nb<-1000; tb<-numeric(nb); predb<-numeric(nb)
thetab<-numeric(nb)
for(i in 1:nb){
  rb<-sample(res,n,replace=T)
  yb<-c1* (1-exp(-c2*x))+rb
  ajustb<- nls(yb~ b1* (1-exp(-b2*x)),start = list(b1= 2, b2=0.001))
  thetab[i]<-summary(ajustb)$coefficients[1]
  sdthetab<-summary(ajustb)$coefficients[3]
  predb[i]<- summary(ajustb)$coefficients[1] * (1-exp(-
    summary(ajustb)$coefficients[2] *900))
  tb[i]<-(thetab[i]-c1)/sdthetab}
```

And this is the bootstrap-t 95% CI of b_1 :

`c1+sdtheta*quantile(tb,0.025)`

231.0746

`c1+sdtheta*quantile(tb,0.975)`

243.6187

This is the CI based in the quantile method,

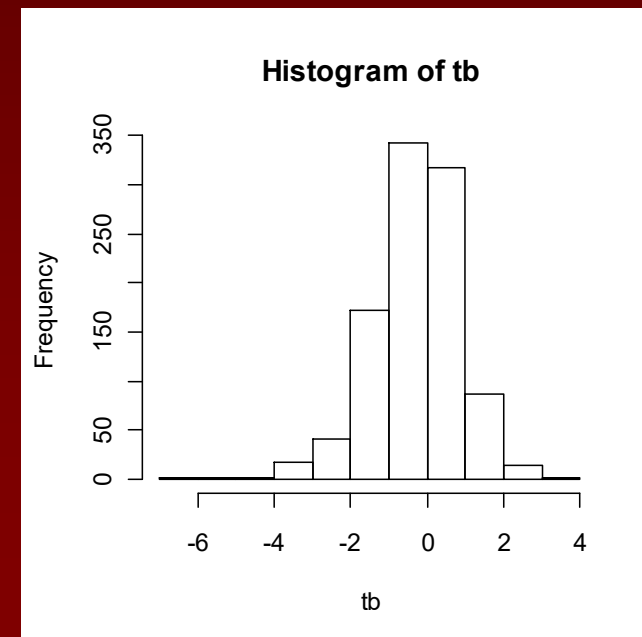
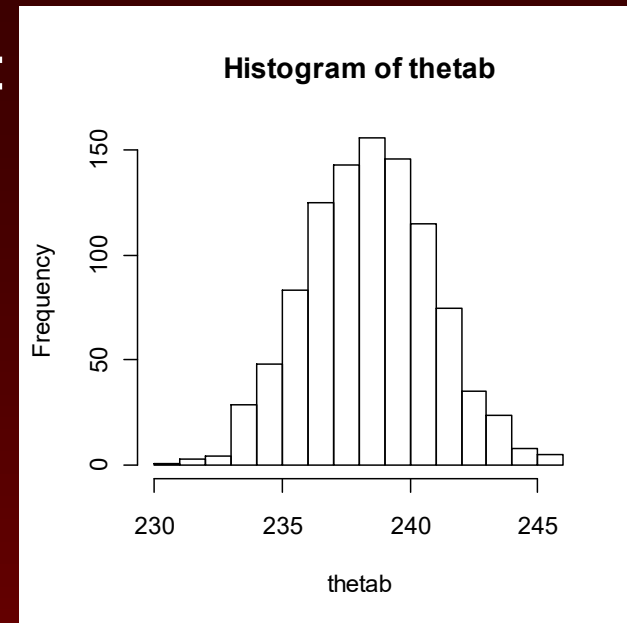
`quantile(thetab,c(0.025,0.975))`

2.5% 97.5%

233.7229 243.4214

And again the classical CI is,

238.9 ± 5.3

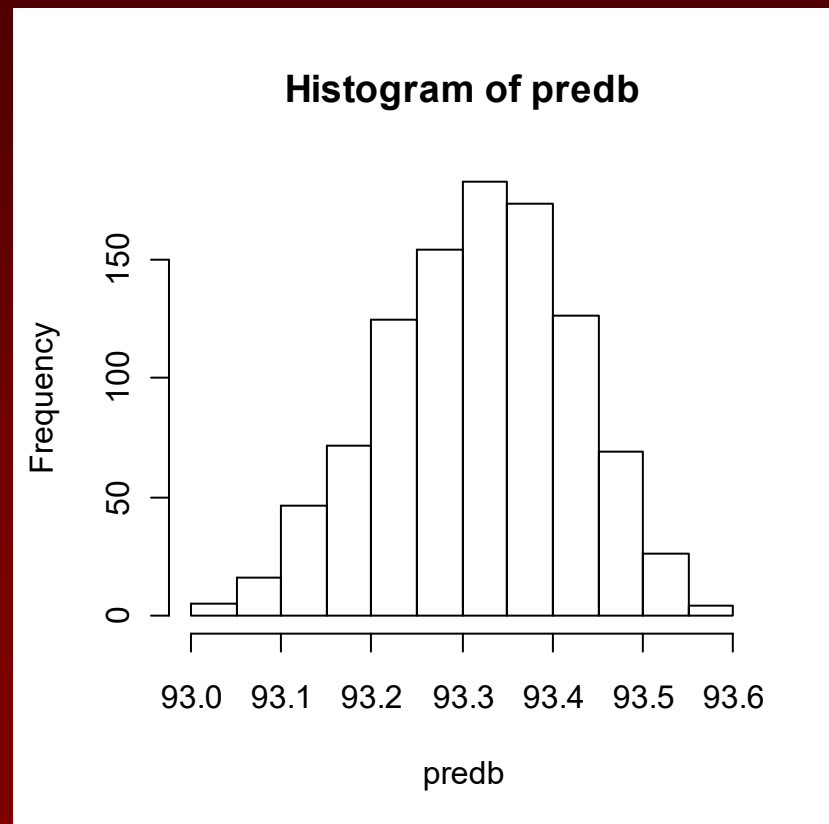


This is the quantile 95% CI of the prediction at x=900:

```
quantile(predb,c(0.025,0.975))
```

2.5% 97.5%

93.10760 93.51087



The bootstrap method in AR(p) time series

The AR(p) model. Is a process described by the expression,
$$Y_t = \Phi_0 + \Phi_1 Y_{t-1} + \Phi_2 Y_{t-2} + \dots + \Phi_p Y_{t-p} + Z_t,$$
where Z_t is a white noise, and all the roots of the characteristic polynomial $P(x) = x^p - \Phi_1 x^{p-1} - \dots - \Phi_p$ are inside the unit circle.

Due to the stationarity condition we can write that $E(Y_t) = \mu$, and μ can be estimated by the sample mean.

Moreover, it can be shown that $X_t = Y_t - \mu$ is an AR(p) process without constant:

$$X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \dots + \Phi_p X_{t-p} + Z_t$$

The ACF of an AR(p) model satisfies the recurrence relation,

$$\rho(s) = \sum_{i=1}^p \Phi_i \rho(s-i) \quad \text{for } s=1,2,\dots$$

The ACF can be estimated by using the correlogram or empirical ACF. Given the observations $\{y_t\}$ of a time series, the k -order empirical correlation coefficient, $r(k)$, is defined as

$$\bar{y} = \frac{1}{n} \sum_{t=1}^n y_t$$

$$c(k) := \frac{1}{n} \sum_{t=1}^{n-k} (y_{t+k} - \bar{y})(y_t - \bar{y})$$

$$r(k) := \frac{c(k)}{c(0)} = \frac{\sum_{t=1}^{n-k} (y_{t+k} - \bar{y})(y_t - \bar{y})}{\sum_{t=1}^n (y_t - \bar{y})^2}$$

The plot of this sample autocorrelations $r(k)$, versus k (the time lags) is called *correlogram*.

The consequences of Yule-Walker equations are important.
The equations can be written in matrix form:

$$\begin{pmatrix} 1 & \rho_1 & \cdots & \rho_{p-1} \\ \rho_1 & 1 & \cdots & \rho_{p-2} \\ \vdots & & & \vdots \\ \rho_{p-1} & \rho_{p-2} & \cdots & 1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_p \end{pmatrix}$$

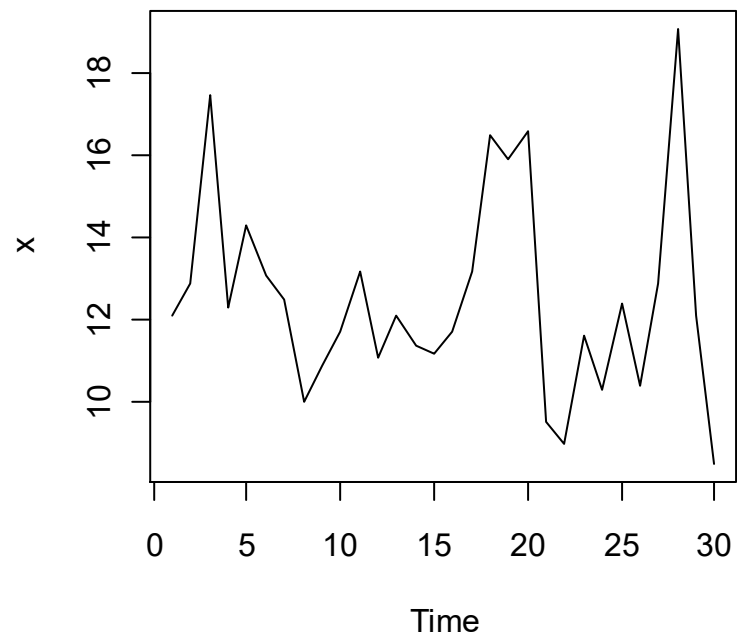
Therefore, given a set of observations coming from an AR(p) model, if we have a way to estimate the ρ_s we can also estimate the Φ_i coefficients.

But these estimates could be the empirical correlation coefficients (correlogram). This is a non-parametric approach. It does not matter what is the distribution of the white noise.

Consider the following time series with 30 observations:

```
x<-c(12.1,12.9,17.5,12.3,14.3,13.1,12.5,10,10.9,11.7,13.2,  
11.1,12.1,11.4,11.2,11.7,13.2,16.5,15.9,16.6,9.5,9,11.6,10.3,  
12.4,10.4,12.9,19.1,12.1,8.5)  
ts.plot(x)
```

We want to fit it using
an AR(2) model.



```
fit<-ar(x, aic = FALSE, order.max = 2,method=c("yule-walker"))  
fit
```

Coefficients:

1	2
0.2872	-0.2091

Order selected 2 sigma^2 estimated as 6.035

```
mean(x)
```

```
[1] 12.53333
```

The estimated process is

$$Y_t - 12.53 = 0.287 (Y_{t-1} - 12.53) - 0.209(Y_{t-2} - 12.53) + Z_t$$

$$Y_t = 11.554 + 0.287 Y_{t-1} - 0.209 Y_{t-2} + Z_t$$

Bootstrapping the residuals

- 1- Fit the original time series and keep the residuals and the estimates of the parameters.

```
res<-fit$resid
```

```
res
```

```
[1]      NA      NA 4.77074255 -1.58291830 2.87241982 0.01053517  
[7] 0.17342500 -2.40524627 -0.91281086 -0.89412195 0.56437248 -  
    1.79906554  
[13] 0.11770422 -1.30866624 -1.09850447 -0.68747134 0.62711559  
    3.60093446  
[19] 2.36699347 3.92946893 -3.49703536 -1.81173729 -0.55307336 -  
    2.70428372  
[25] 0.31280920 -2.56213182 0.95140731 6.01519823 -2.24238815 -  
    2.53551650
```

Note that the two first values are missing.

2- Construct the bootstrap series using the estimated process and a sample of the residuals instead of Z_t ,

$$YP_t = 11.554 + 0.287 YP_{t-1} - 0.209 YP_{t-2} + \text{resp}_t$$

The two first values of the bootstrap series are the same than the original series.

```
nb<-1000; mu<-numeric(nb); phi1<-numeric(nb)
phi2<-numeric(nb)
xb<-numeric(30)

for(j in 1:nb){
  xb[1]<-x[1];xb[2]<-x[2]
  for (i in 3:30){
    r<-sample(res[3:30],1,replace=T)
    xb[i]<-fit$ar[1]*xb[i-1]+fit$ar[2]*xb[i-2]+mean(x)*(1-fit$ar[1]-
      fit$ar[2])+r}
  fitb<-ar(xb, aic = FALSE, order.max = 2,method=c("yule-walker"))
  mu[j]<-mean(xb);phi1[j]<-fitb$ar[1]; phi2[j]<-fitb$ar[2] }
```

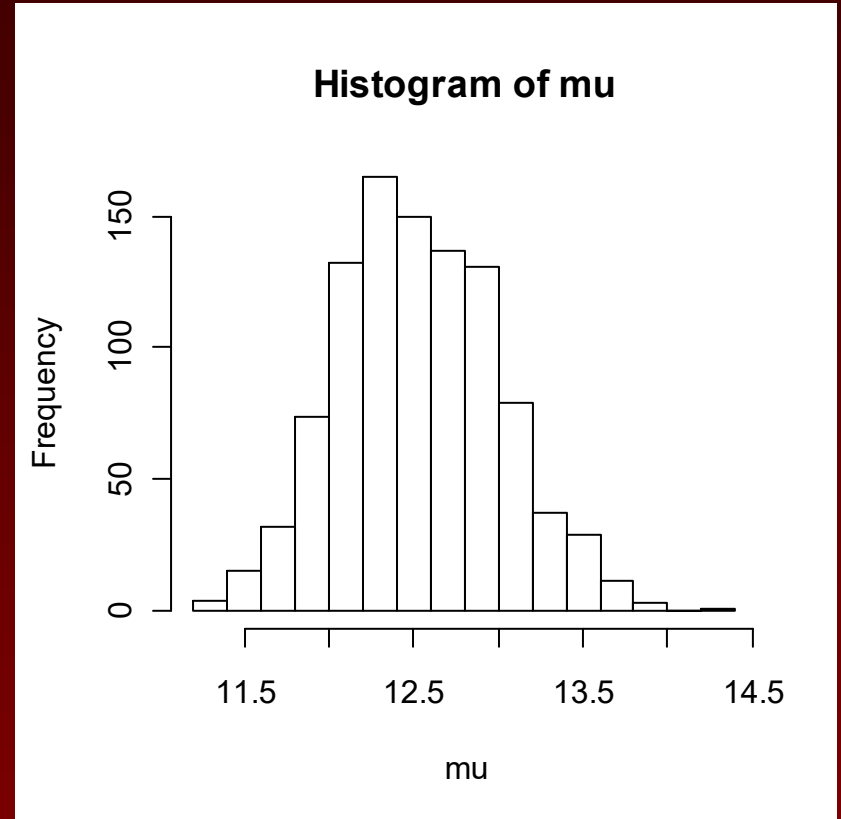
This is the bootstrap 95% CI of μ ,
based in quantiles:

```
quantile(mu,c(0.025,0.975))
```

2.5% 97.5%

11.64771 13.50074

This is, 12.57 ± 0.93



These are the quantile 95% CI of the coefficients:

`quantile(phi1,c(0.025,0.975))`

2.5% 97.5%

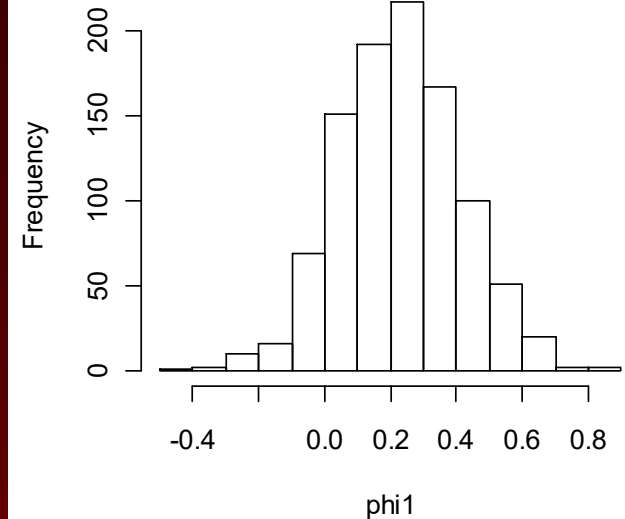
-0.1260521 0.5834652

`quantile(phi2,c(0.025,0.975))`

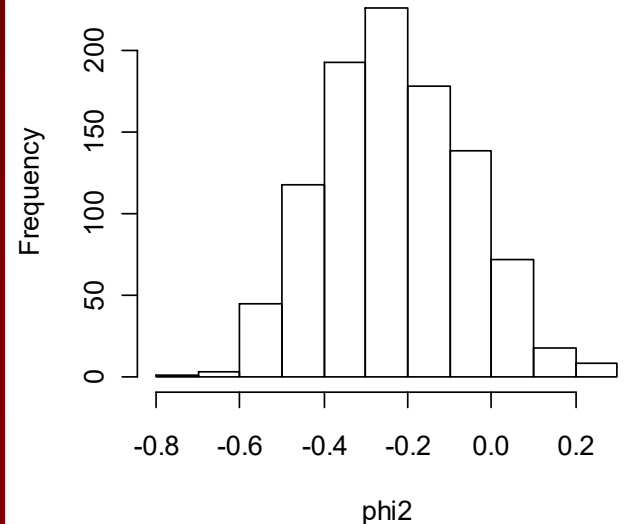
2.5% 97.5%

-0.5393872 0.1033358

Histogram of phi1



Histogram of phi2



Litvinenko case



Natasja Weitsz / Getty

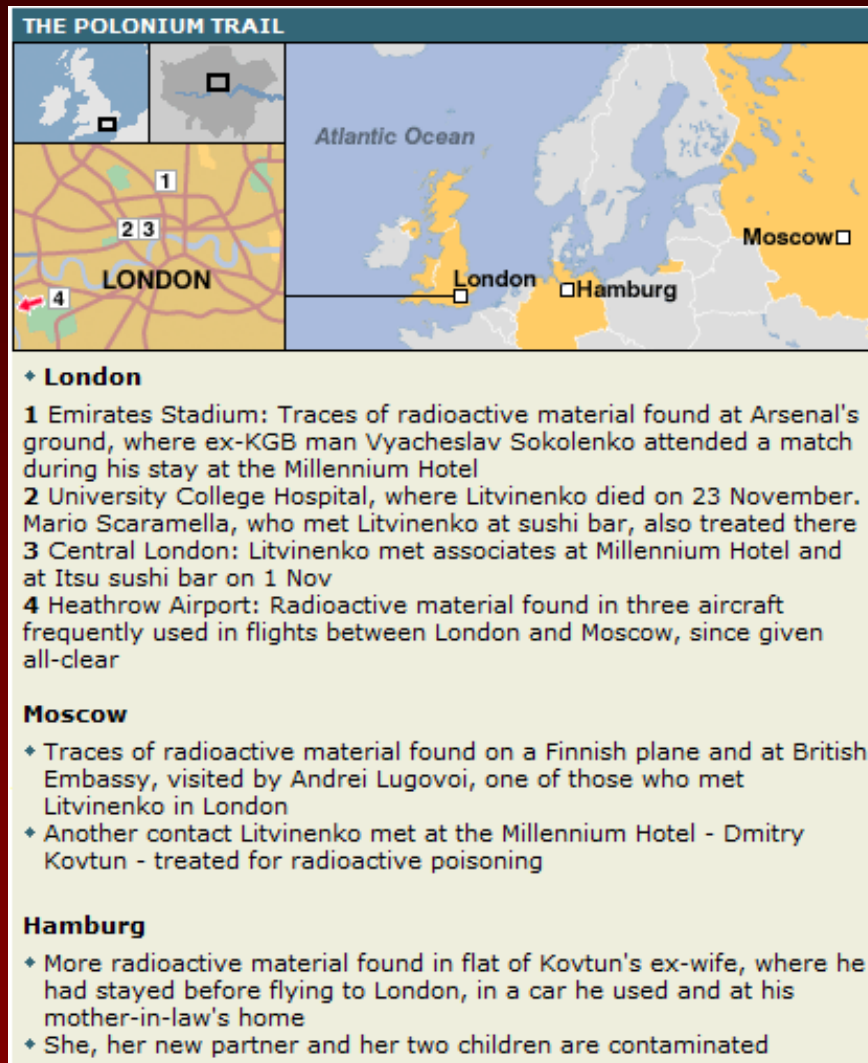
Alexander Litvinenko in the intensive care unit of University College London Hospital on November 20, 2006, three days before his death from radiation poisoning.

The mysterious circumstances surrounding the death of Litvinenko, apparently from radiation poisoning, spawned an international crisis. Britain demanded that Russia extradite a Russian citizen allegedly connected to the

case. When it refused, Britain expelled four Russian diplomats from London, in reprisals reminiscent of the Cold War.

<https://www.youtube.com/watch?v=KpReRcw3mf0>

Litvinenko case



More than 150 persons were examined to detect possible radioactive contamination.

There are many radiation accidents

- Radiotherapy accidents
- Manipulation of radioactive components in laboratories
- Industrial radiographies
- Disasters (Chernobyl (1986), Fukushima (2011))



From: Igor Kostin

There are many radiation accidents

Acerinox accident

From Wikipedia, the free encyclopedia

The **Acerinox accident** was an incident of [radioactive contamination](#) in [Cádiz \(Spain\)](#). In May 1998, a [caesium-137](#) source managed to pass through the monitoring equipment in an [Acerinox scrap metal](#) reprocessing plant in [Los Barrios](#), Spain. When melted, the caesium-137 caused the release of a [radioactive cloud](#). The Acerinox [chimney](#) detectors failed to detect it, but it was eventually detected in France, Italy, Switzerland, Germany and Austria. The [radioactive](#) levels measured were up to 1000 times higher than normal.

The accident contaminated the [scrap metal](#) reprocessing plant, plus two other steel mills where it sent its waste for decontamination. According to independent laboratories,^[1] the ashes produced by the Acerinox factory had between 640 and 1420 [becquerels](#) per gram (the [Euratom](#) norm is 10 Bq/g), high enough to be a threat to the public.

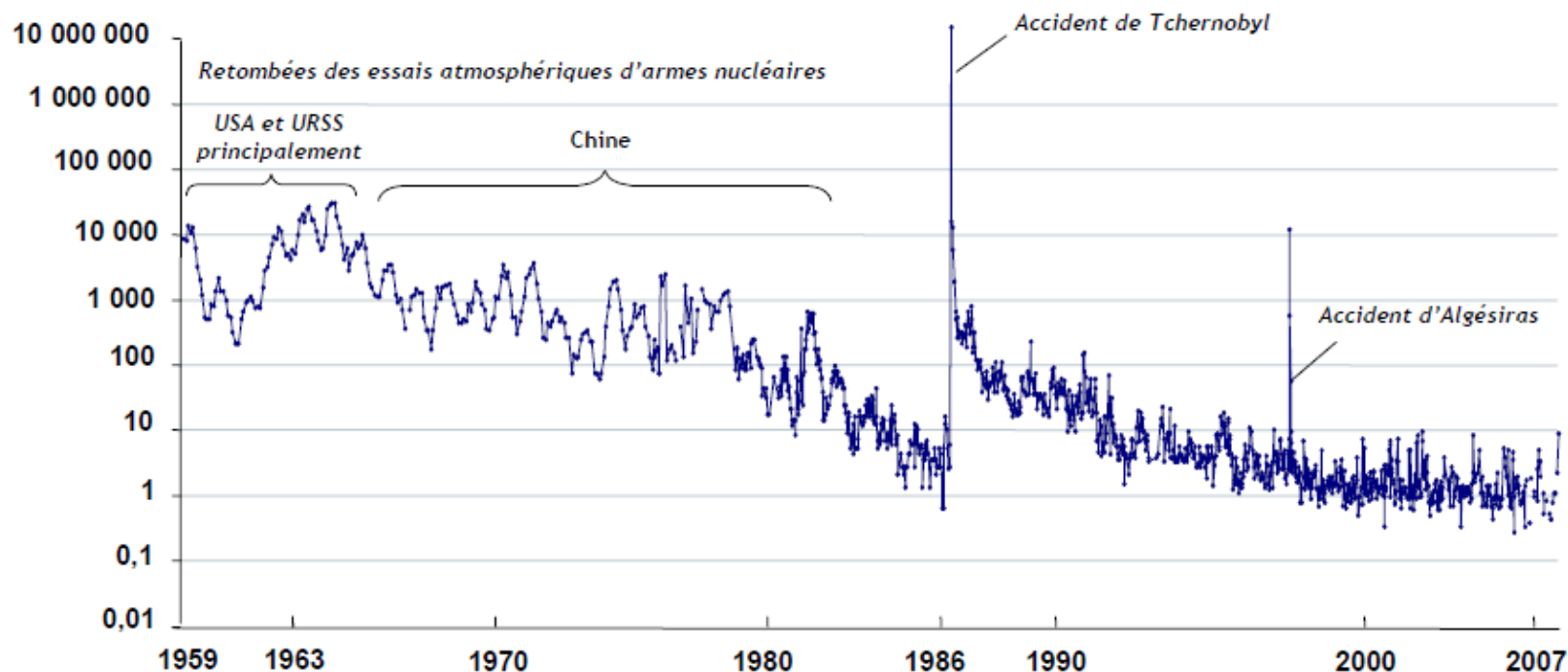
On the radiological consequences of this event, six people were exposed to slight levels of caesium-137 contamination. The estimated total costs for clean-up, waste storage, and lost production in the factory were around 26 million US dollars (most of it due to the lost production).^[2]

WIKIPEDIA
The Free Encyclopedia



Historique des mesures de césium 137 dans l'air en France depuis le début des années 60

Activité volumique du ^{137}Cs dans les aérosols $\mu\text{Bq}/\text{m}^3$ d'air



Station OPERA de la Seyne-sur-Mer

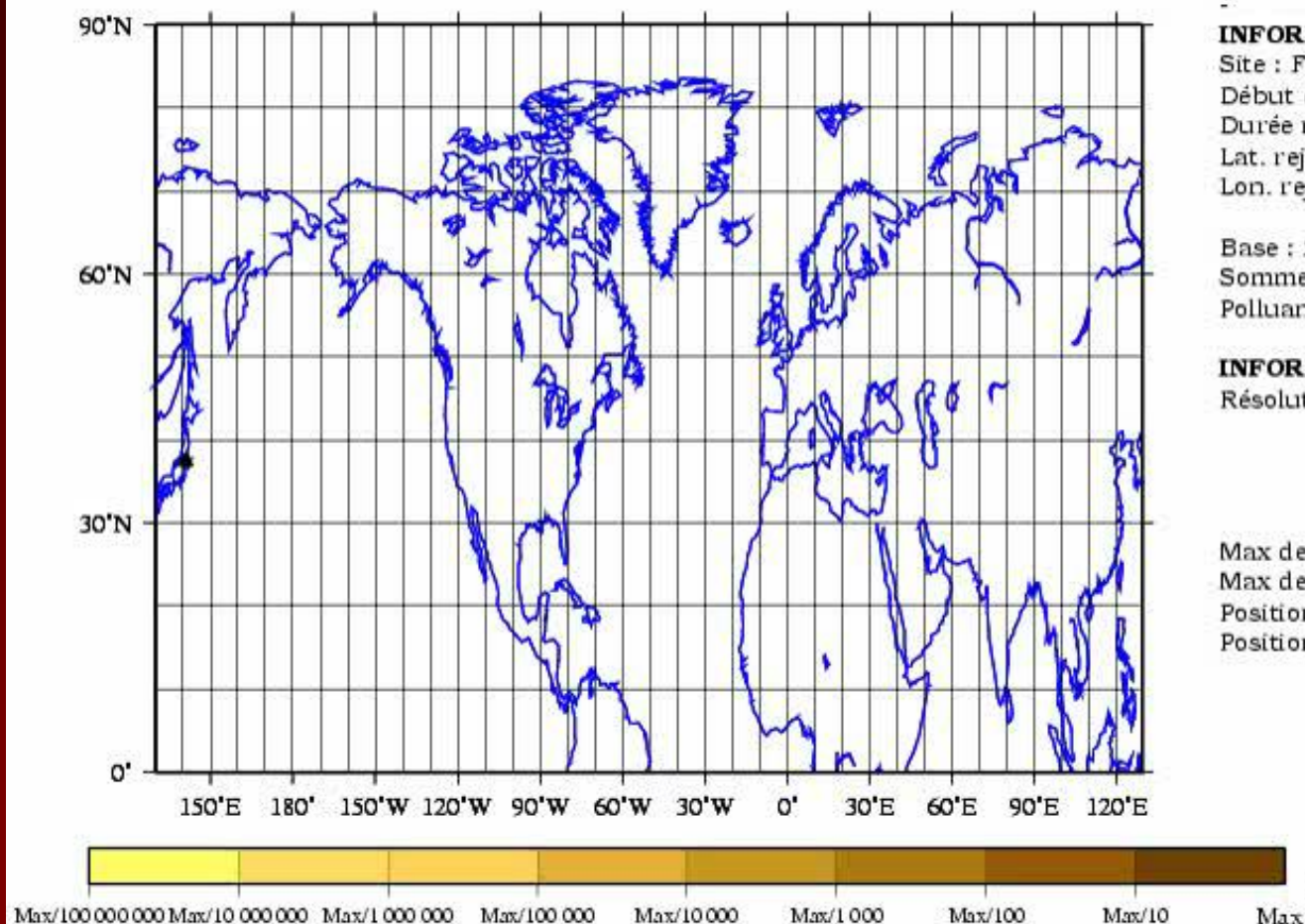
Fukushima's accident



US Government Report Declassified: The aftermath of the Fukushima Disaster

Accident de Fukushima_IRSN du 11/03/2011

Concentration moyenne sur la couche 0m-500m en bq/m³
11/03/2011 12h00 UTC



Modèle de dispersion :
MOCAGE Accident
Modèle météo : CEP

INFORMATIONS REJET

Site : Fukushima_IRSN
Début émission : 11/03/2011 11h UTC
Durée rejet : scénario spécifié par l'IRSN
Lat. rejet : 37.421 N : 37°25'15" N
Lon. rejet : 141.033 E : 141°1'58" E

Base : 20m
Sommet : 500m
Polluant émis : Cs-137

INFORMATIONS MODELE

Résolution de la grille : 0.5°

Max de l'échéance = 1.3e-03

Max de l'échelle = 1.0e+03

Position du maximum ★

Position de la source ▲



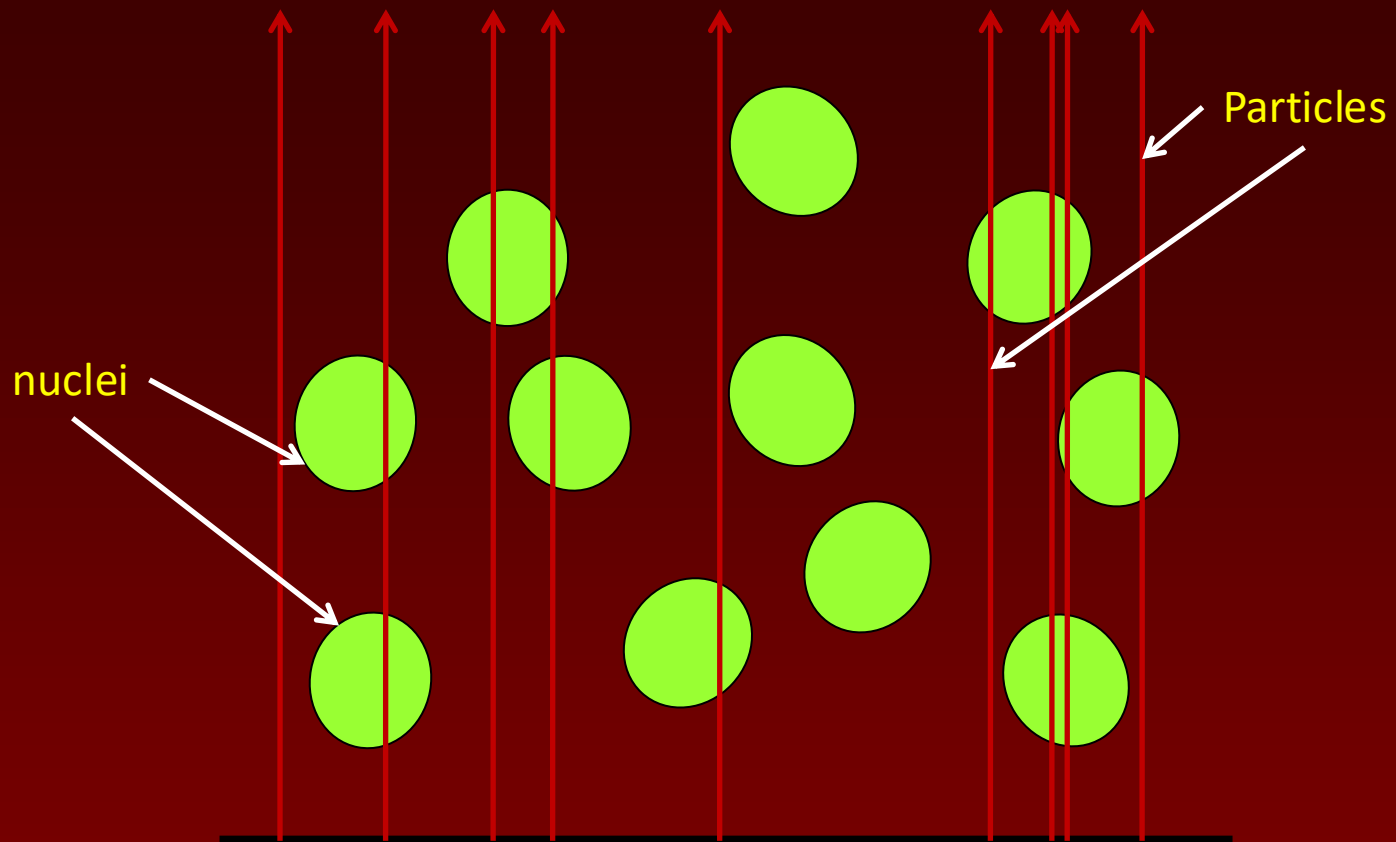
CMRS TOULOUSE

Créé le 05/04/2011 16:21:47 UTC

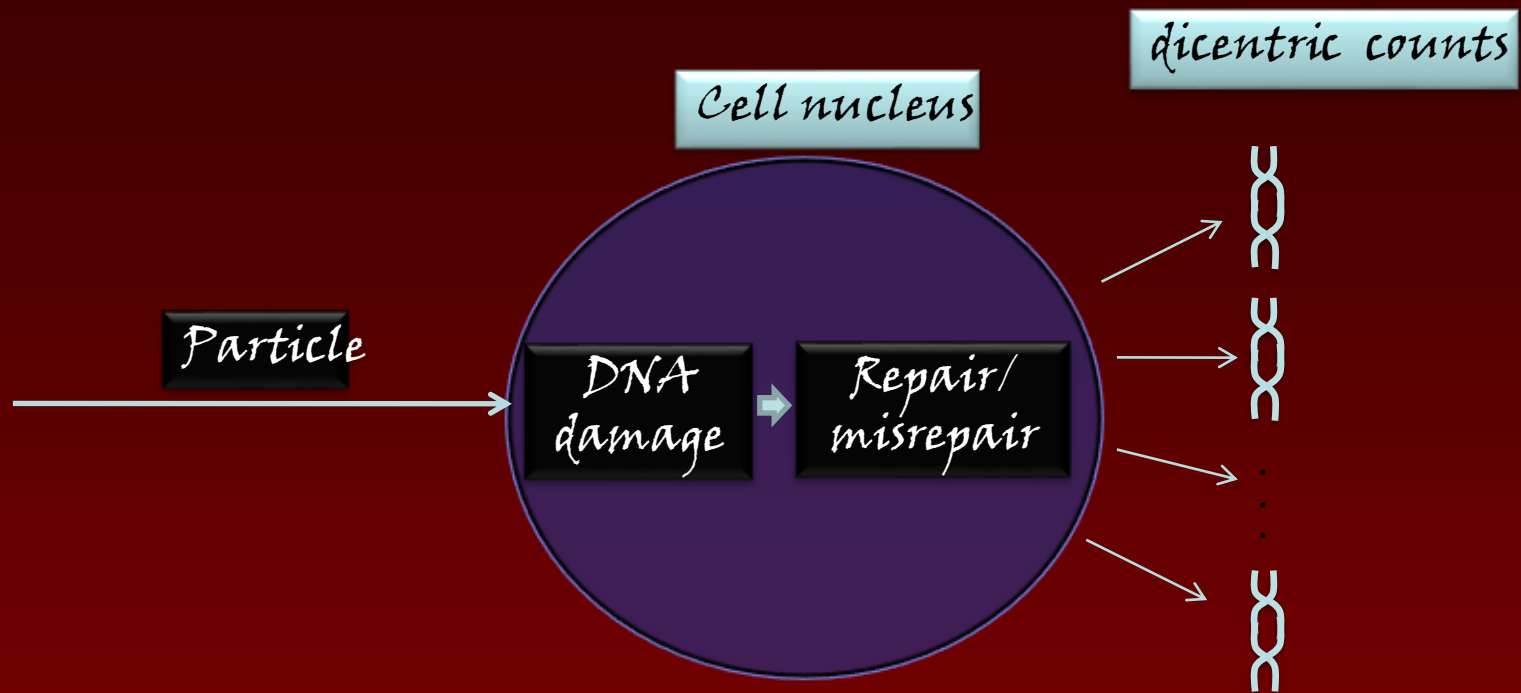
Quantification of the radiation dose received

This is essential for predicting the derived health consequences, for both early effects (vomiting, skin Injuries, etc.) and late effects like the development of cancer.

The most widely used method is the analysis of the frequency of dicentrics observed in metaphases from in peripheral blood lymphocytes.



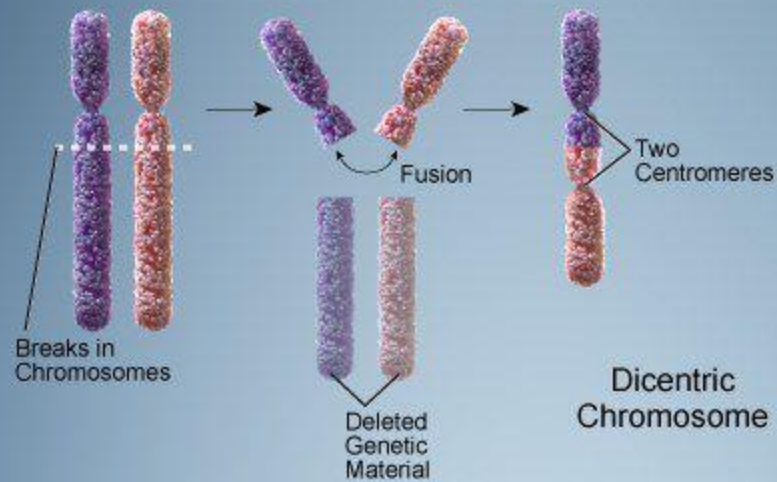
Poisson-distributed events. The figure shows events, one event is a track (in red) intersecting a cell nucleus (green)



Dicentric formation as a Compound Poisson process



Uniformly stained metaphase spread with dicentric chromosomes (arrows).



U.S. National Library of Medicine



Dicentric chromosome generation due to the breaks produced by radioactive particles

Natural radiation also exists

Banana equivalent dose

From Wikipedia, the free encyclopedia

A **banana equivalent dose (BED)** is an informal expression of [ionizing radiation](#) exposure, intended as a general educational example to indicate the potential dose due to naturally occurring [radioactive isotopes](#) by eating one average-sized [banana](#). One BED is often taken as $0.1\ \mu\text{Sv}$, however, in practice this dose is not [cumulative](#), as the principal radioactive component is excreted to maintain [metabolic equilibrium](#). The BED is only an indicative concept meant to show the existence of very low levels of natural radioactivity within a natural food and is not a formally adopted dose quantity.

For example, the radiation exposure from consuming a banana is approximately 1% of the average daily exposure to radiation, which is 100 banana equivalent doses (BED). The maximum permitted radiation leakage for a nuclear power plant is equivalent to 2,500 BED (250 μSv) per year, while a chest [CT scan](#) delivers 70,000 BED (7 mSv). A [lethal dose](#) of radiation is approximately 35,000,000 BED (3500 mSv). A person living 10 miles from the [Three Mile Island](#) nuclear reactor received an average of 800 BED of exposure to radiation during the 1979 accident.^[1]



A banana contains [naturally occurring radioactive material](#) in the form of [Potassium-40](#).

The main goal: what is the dose received ?



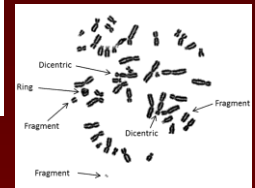
Irradiated patient



Blood sample



Cytogenetic methods

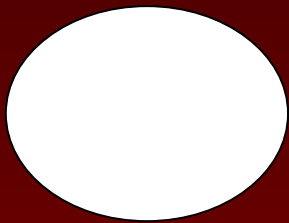


Frequencies of chromosome aberrations in peripheral lymphocytes

OBSERVED	cells with X dicentrics									CELLS	DIC	Freq
	0	1	2	3	4	5	6	7	8			
224	180	72	19	4	1	0	0	0	0	500	402	0.80

How the aberrations are distributed among cells

For dicentrics, irradiation with X or gamma rays produces a distribution of damage which is very well represented by the Poisson distribution.



$$\frac{\lambda^0 e^{-\lambda}}{0!}$$



$$\frac{\lambda^1 e^{-\lambda}}{1!}$$



$$\frac{\lambda^2 e^{-\lambda}}{2!}$$



$$\frac{\lambda^3 e^{-\lambda}}{3!}$$



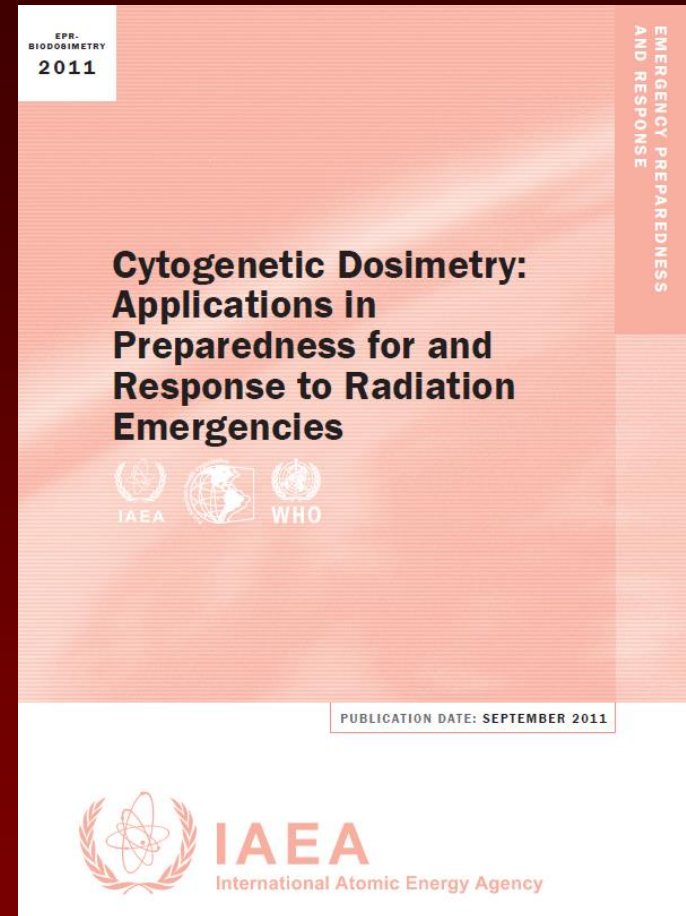
$$\frac{\lambda^4 e^{-\lambda}}{4!}$$

λ is related to the dose received: it increases when the doses increases

According to the IAEA manual:

The Poisson distribution is by far the most widely recognized and commonly used type of distribution for cytogenetic data analysis. (p. 197)

In contrast, neutrons and other types of high LET radiation produce distributions which display patterns not compatible with Poisson distribution.



An application of compound Poisson modelling to biological dosimetry

Pedro Puig and Joan Francesc Barquinero

Proc. R. Soc. A published online 29 September 2010
doi: 10.1098/rspa.2010.0384



F. Barquinero (UAB and IRSN)

Bayesian methods

Mutation Research 756 (2013) 184–191



Contents lists available at ScienceDirect

Mutation Research/Genetic Toxicology and Environmental Mutagenesis

journal homepage: www.elsevier.com/locate/gentox

Community address: www.elsevier.com/locate/mutres



CytoBayesJ: Software tools for Bayesian analysis of cytogenetic radiation dosimetry data



Elizabeth A. Ainsbury^{a,*}, Volodymyr Vinnikov^{b,1}, Pedro Puig^{c,2}, Nataliya Maznyk^{b,1}, Kai Rothkamm^a, David C. Lloyd^a

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^c Departament de Matemàtiques, Universitat Autònoma de Barcelona, 08193 Cerdanyola del Vallès, Spain

A new Inverse Regression Model Applied to Radiation Biodosimetry

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de Barcelona, Bellaterra, Barcelona 08193, Spain



Higuera, Ainsbury and Rothkamm

The dose-response curve

The first step is to do a calibration, that is, to construct a dose-response curve. The usual approach is to irradiate in the laboratory n blood samples from a healthy donor with several doses. For low LET radiation, 10 or more doses should be used in the range 0.25–5.0 Gy (IAEA, 2011).

The construction of the dose-response curve can be experimentally difficult according to the kind of radiation.

doses (Gy) =	0.5	0.75	1.0	1.5	2.0	2.5	3.0
Number of cells=	269	78	115	90	84	59	37
Number of dicentrics=	109	47	94	114	138	125	97

(Groer and Pereira, 1987)

There is a very strong empirical evidence that the yields (population mean) of chromosome aberrations (Y) are related to dose (D) by the linear-quadratic equation.

$$Y = C + \alpha D + \beta D^2$$

Some times (high LET radiation, for instance) the α -term becomes large and eventually the β -term becomes biologically less relevant and also statistically 'masked' and the dose response is approximated by the linear equation .

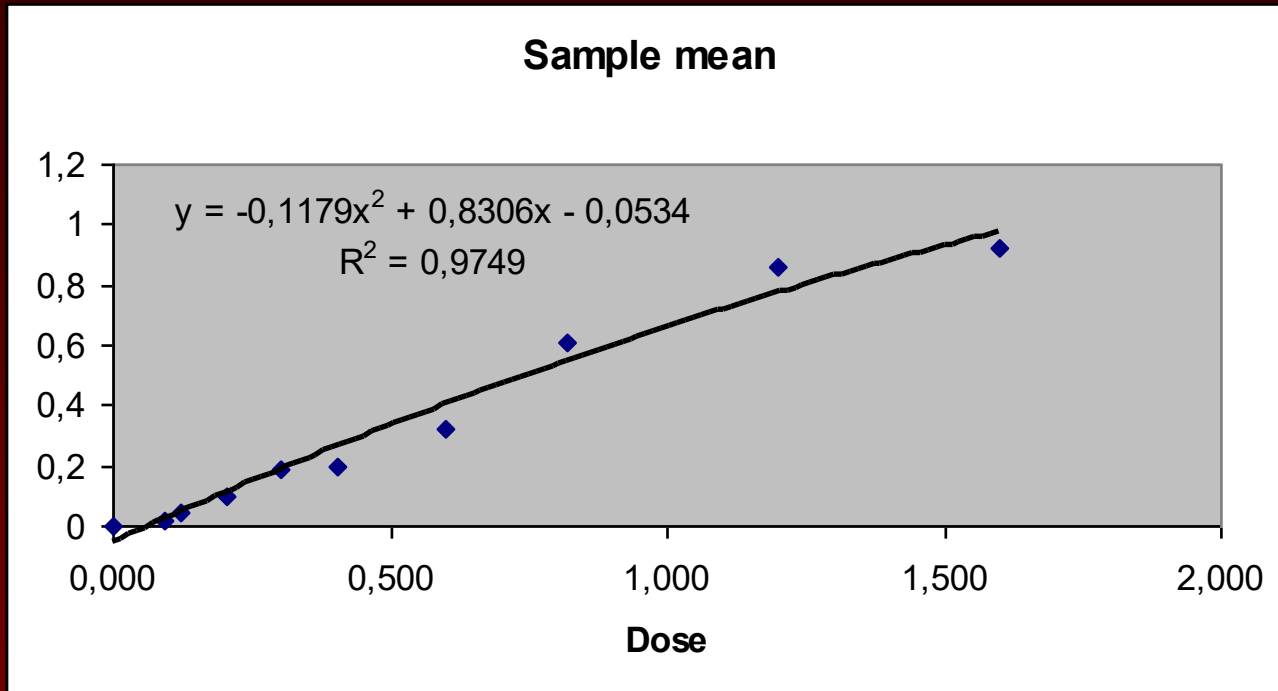
$$Y = C + \alpha D$$

An example

1480 MeV oxygen ions															
dose (Gy)	N	X	cell distribution of dicentrics								dic	mean	var	D	
			0	1	2	3	4	5	6	7					
0,000	2000	1	1999	1	0	0	0	0			1	0,0005	0,0005	1,00	
0,092	753	16	737	16	0	0	0	0			16	0,0212	0,0208	0,98	
0,120	1500	71	1438	55	5	2	0	0			71	0,0473	0,0598	1,26	
0,205	1420	138	1300	104	14	2	0	0			138	0,0972	0,1160	1,19	
0,300	560	106	471	73	15	1	0	0			106	0,1893	0,2181	1,15	
0,405	520	103	437	66	15	1	1	0			103	0,1981	0,2516	1,27	
0,600	631	204	473	119	34	3	2	0	0		204	0,3233	0,3937	1,22	
0,820	413	253	253	99	38	17	5	0	1		253	0,6126	0,9175	1,50	
1,200	190	163	92	55	27	11	4	1			163	0,8579	1,1173	1,30	
1,600	173	160	80	49	26	13	5	0			160	0,9249	1,1746	1,27	

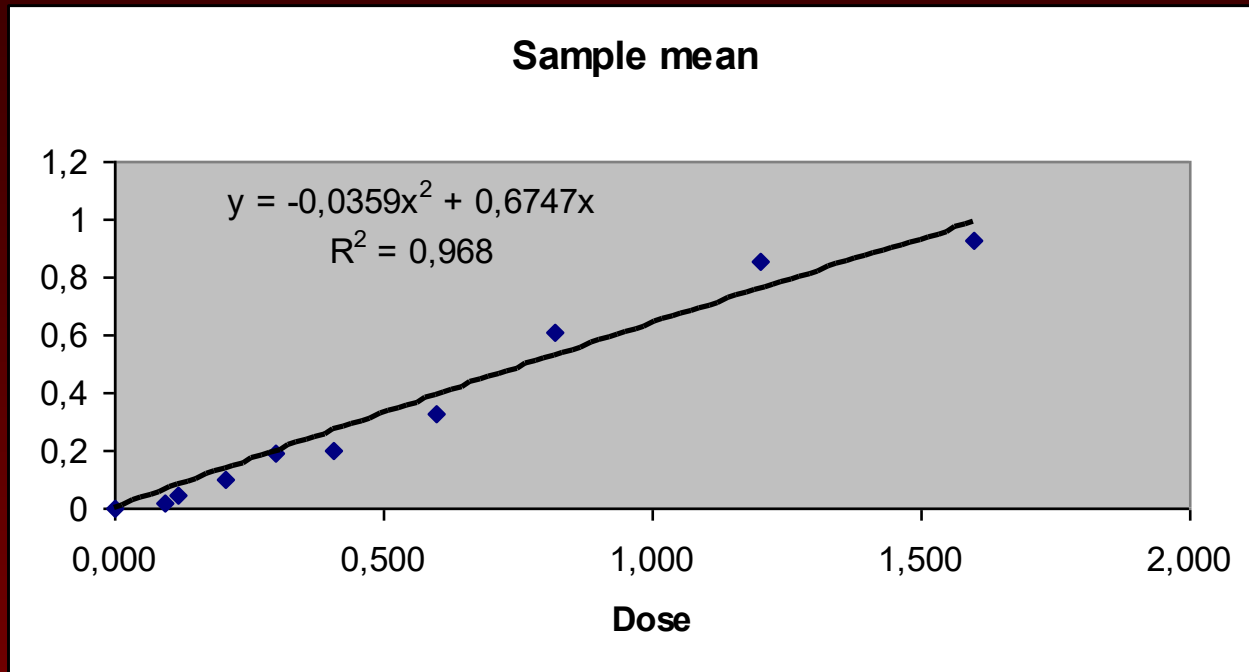
It has been done in order to calibrate the method.
Here the doses are perfectly known.

An example



We can use the OLS techniques and the inverse regression methods but, are they appropriate here? OLS techniques can be understood as a nonparametric approach if the CI are calculated in a nonparametric way.

An example



We can use the OLS techniques (without intercept) and the inverse regression methods but, are they appropriate here? A model without intercept assumes that $y=0$ when $x=0$ and it could be reasonable here.

We want to calculate a 95% confidence interval of the doses of radiation received for a person which a mean of dicentrics equal to 0.7.

We shall consider the method of residuals bootstrapping.

```
dose<-c(0.000,0.092,0.120,0.205,0.300,0.405,0.600,0.820,1.200,1.600)
dm<-c(0.0005,0.0212,0.0473,0.0972,0.1893,0.1981,0.3233,0.6126,0.8579,0.9249)
dose2<-dose^2
fit<-lm(dm ~ dose + dose2 -1)
summary(fit)
```

Coefficients:

	Estimate	Std. Error	t	value	Pr(> t)
dose	0.67468	0.09309	7.248	8.82e-05 ***	
dose2	-0.03588	0.07082	-0.507	0.626	

A point estimate of the received doses is the positive root of the second order equation,

$$0.7 = 0.67468 \cdot x - 0.03588 \cdot x^2$$

```
dmob<-0.7
```

```
a<-summary(fit)$coefficients[2]; b<-summary(fit)$coefficients[1]
```

```
dest<- (-b+sqrt(b*b+4*a*dmob))/(2*a)
```

```
dest
```

```
1.102117
```

Due to the nature of our problem we are going to use the method of bootstrapping the residuals.

```
res<-residuals(fit)
```

```
n<-length(dose)
nb<-1000; destb<-numeric(nb)

for(i in 1:nb){
  rb<-sample(res,n,replace=T)
  dmb<- b*dose+a*dose^2 + rb
  fitb<- lm(dmb ~ dose + dose2 -1)
  anew<-summary(fitb)$coefficients[2];
  bnew<-summary(fitb)$coefficients[1]
  destb[i]<- (-bnew+sqrt(bnew*bnew+4*anew*dmob))/(2*anew)}
```

This is the quantile 95% CI of the doses of radiation received:

```
quantile(destb,c(0.025,0.975))
```

2.5% 97.5%

1.026268 1.211516

It corresponds to, 1.12 ± 0.09

#Simple method:

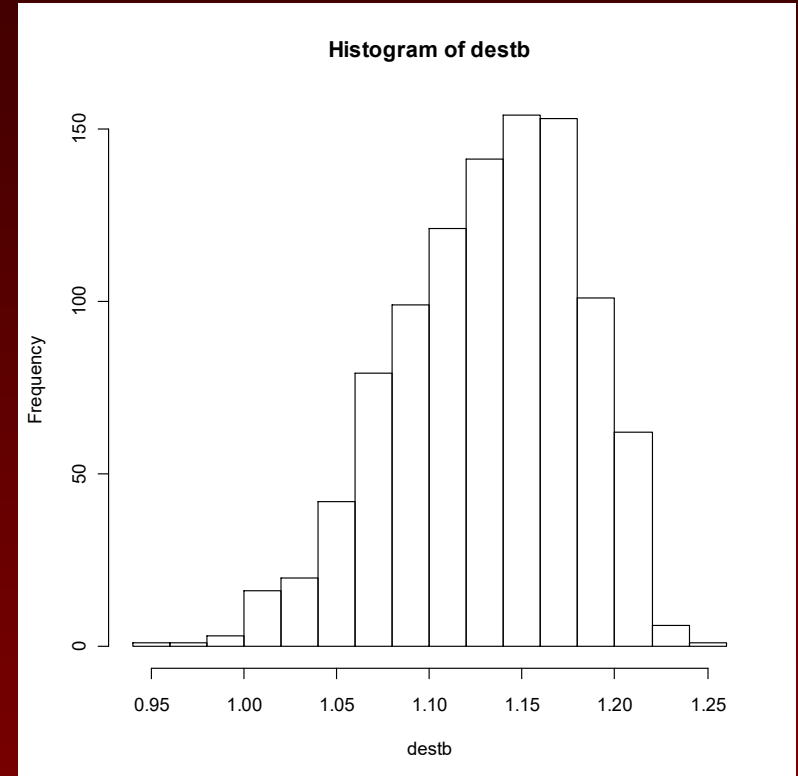
```
2*(dest)-quantile(destb,0.975)
```

0.9927167

```
2*(dest)-quantile(destb,0.025)
```

1.177965

It corresponds to, 1.085 ± 0.092



Radiation Cytogenetics Database

<http://www.rbc.kyoto-u.ac.jp/db/radchrom.html>

The IAEA manual

<http://www-pub.iaea.org/books/IAEABooks/Series/124/Emergency-Preparedness-and-Response>